Eclipse-gated transient energy gain for passive solar sails in near-polar low Earth orbit

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Abstract

Solar sails in low Earth orbit are normally expected to lose energy to atmospheric drag. This paper asks whether a passive, spin-stabilised sail can instead gain orbital energy, and shows that it can — but only briefly, and only within a ~~narrow~~ bounded range of orbit geometries. Of ~~900 cases sampled at 700 km, about one in ten gained altitude~~ 1776 orbit-attitude combinations sampled at 800 km, one in five gained mean semi-major axis over two weeks, the best by ~~roughly 20 km~~5.95 km; at 700 km, with the same sail, none did.

Eclipse is the reason. Over each orbit, sunlight pushes the sail in ways that both add and remove energy. When the Earth's shadow covers the part of the orbit where the push is unfavourable, the balance over a full orbit turns positive; remove eclipse from the same case and the orbit decays instead.

The gain does not last. As the orbit stretches, its low point drops deeper into the atmosphere, ~~drag grows, and within about two and a half months the orbit falls below~~ and the fixed sail slowly turns away from the Sun as the seasons advance. The best case peaks 10.9 km above its starting orbit after five weeks and is back where it began after ten.

To show that the effect is physical rather than an artefact of one implementation, the propagator used here is released openly and cross-checked against Ansys STK/Astrogator and Orekit~~, both of which reproduce it. The choice of atmospheric~~ ; both reproduce the eclipse-gated gain and the drag decay once two configuration errors in the original STK set-up were corrected. The thermospheric density model, not the sail physics, dominates the remaining uncertainty.

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1. Introduction

Solar radiation pressure (SRP) provides a continuous, propellant-free means of modifying spacecraft orbits, and has been demonstrated in missions including IKAROS [1, 2], LightSail 2 [3], and the Advanced Composite Solar Sail System [4]. Recent reviews summarise the current state of solar sailing technology and mission heritage [5, 6]. While active solar sailing enables controlled trajectory shaping, ~~the~~ and articulated vanes, reflectivity-control devices or displaced masses give modern sails considerable attitude authority [7], a passive spin-stabilised configuration is a far simpler alternative that forgoes any control over the long-term orbital-energy dynamics. The possibility of achieving meaningful orbital energy change under such purely passive configurations remains less well understood, particularly in low Earth orbit (LEO) where atmospheric drag and gravitational perturbations are significant. The use of solar sails in LEO has been explored for deorbiting [8, 9] and for altitude maintenance [10], but the short-horizon interaction between eclipse timing and SRP work balance has received limited attention.

Prior work has examined the coupled effects of solar radiation pressure and atmospheric drag on high area-to-mass spacecraft. In particular, Colombo and McInnes [11] demonstrated that asymmetric SRP, including the influence of Earth eclipse, can be exploited to balance drag losses and produce long-lived Earth-centred orbits without active control. The influence of SRP on high area-to-mass ratio objects has also been studied in the context of long-term orbit evolution and resonance structures [12]. These analyses highlight the importance of SRP–drag interaction and shadow-induced asymmetries in shaping orbital evolution. However, much of the existing SRP–drag literature emphasizes long-term or averaged behaviour and equilibrium-like manifolds or "balance" conditions, rather than isolating short-horizon regimes in which the timing of eclipse arcs alters the sign and magnitude of orbit-averaged SRP work.

In near-polar LEO, the interaction between SRP, atmospheric drag, Earth's oblateness (J_2), and eclipse timing produces a dynamically coupled environment in which small perturbations can lead to non-intuitive orbital behaviour over short timescales [13]. In particular, a passively oriented, spin-stabilised

sail experiences alternating phases of positive and negative SRP work over each orbit, which would be expected to largely cancel in the absence of asymmetries [14].

However, eclipse introduces a temporal modulation of SRP by removing solar forcing over specific orbital arcs [15]. Depending on the orbital geometry, this can selectively suppress phases in which SRP performs negative work, altering the orbit-averaged energy balance. This raises the question of whether a net increase in semi-major axis can arise over a bounded time horizon under restricted conditions, without active attitude control.

This work investigates that question by isolating and characterising a transient, geometry-dependent energy-pumping regime in near-polar LEO for spin-stabilised solar sails. The aim is not to propose a mission architecture or demonstrate sustained orbit raising, but to identify where the regime exists, explain the eclipse-modulated SRP work-imbalance mechanism, and quantify the conditions under which it appears and breaks down. Using a ~~reduced-order~~ three-dimensional J_2–drag–SRP dynamical model and parameter-space sweeps over local time of ~~descending node (LTDN~~ascending node (LTAN) and sail orientation, we show that short-horizon semi-major-axis growth can occur within a ~~narrow alignment envelope~~ bounded envelope of sail attitudes and altitudes, but that the effect is inherently self-limiting through coupled eccentricity growth, increased drag exposure near perigee, and ~~gradual drift in Sun-relative geometry~~the drift of the fixed sail normal away from the Sun.

Because the regime is narrow and rests on a fine balance between eclipse-gated SRP work and atmospheric drag, its credibility depends on being reproducible in more than one propagator and on the propagator itself being verified against a trusted reference. This is not merely a matter of numerical hygiene. LEO has become the pre-eminent low-cost testbed for maturing solar- and drag-sail technology: demonstrators such as NanoSail-D [16], CubeSail [17], InflateSail [18], and CanX-7 [19] have all flown on CubeSat-class platforms precisely because they provide rapid, inexpensive access to space for technology maturation [20, 21], and dedicated studies of sail dynamics in LEO have highlighted the strong and competing roles of drag and SRP in this environment [22, 23]. Predicting whether and where a passive sail can gain energy in this environment therefore requires force-model fidelity in exactly the two effects that this regime hinges on, and tools whose

behaviour can be inspected and reproduced by other groups.

Accordingly, this paper has two coupled objectives. The first is scientific: to *verify the existence* of the passive-gain region and characterise its mechanism and limits. The second is methodological: to *validate ~~an~~ the open-source propagator* — built specifically for this class of passive-sail SRP–drag problem, and used for every result in this paper — against two independent references: Ansys Systems Tool Kit (STK) with the Astrogator module [24, 25], an industry-standard ~~, high-fidelity~~ flight-dynamics environment routinely used as a verification reference for orbit-propagation software [26, 27], and the Orekit library [28], which brings its own planetary ephemerides, attitude handling, panel-based force models and adaptive integration. Open-source propagators such as GMAT [27], Orekit~~[28]~~, poliastro [29], Basilisk [30] and Tudat [31] make such analyses transparent and repeatable, but only once their force models are shown to agree with a recognised benchmark. We therefore present the ~~reduced-order~~ analysis of the gain regime (Sections 3–8) together with ~~an independent open-source re-implementation and~~ a component-by-component cross-validation ~~against STK~~ of the propagator that produced it (Section 9). The ~~validation both~~ cross-validation confirms that the ~~gain~~ regime is not an artefact of a single implementation~~and, through the observed inter-tool differences, sharpens the physical interpretation by isolating atmospheric drag~~, and the comparison of atmosphere models identifies thermospheric density — not the sail physics — as the dominant ~~modelling uncertainty for realising the effect~~uncertainty in where the regime is realisable.

2. Scope and Problem Definition

This work is framed as a bounded orbital-dynamics study of a specific short-horizon regime in near-polar LEO in which solar radiation pressure (SRP) can, under restricted passive conditions, produce a net orbit-averaged energy gain. The system considered is intentionally limited to the coupled effects of central gravity, Earth's oblateness (J_2), atmospheric drag, solar radiation pressure, and eclipse-modulated shadowing, for a spin-stabilised sail whose orientation is fixed in inertial space and not actively controlled.

The primary question is whether eclipse timing can, over a bounded horizon, produce a net positive orbit-averaged energy input by selectively suppressing phases of negative SRP work, and how strongly that effect depends on

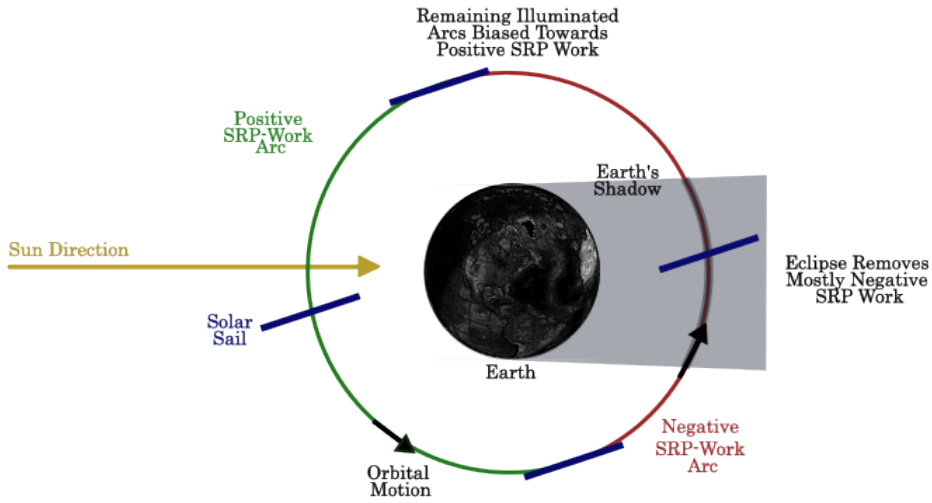


Figure 1: Conceptual schematic of the eclipse-modulated SRP work asymmetry in the reduced mechanism picture. For an inertially fixed sail, solar radiation pressure contributes positive work over part of the illuminated orbit and negative work over another part. If eclipse removes a larger fraction of the unfavourable arc than of the favourable arc, the one-orbit SRP work integral becomes positive. The figure is intended to illustrate the phase-selective gating mechanism, not to represent the full propagated 3D geometry directly.

orbital geometry. The purpose is therefore to isolate and explain a geometric dynamical mechanism, not to demonstrate an operational orbit-raising strategy.

The study is framed around two explicit objectives. The first (scientific) objective is to *verify the existence* of the passive-gain region: to map the constrained dynamical regime, explain its underlying eclipse-modulated SRP work-imbalance mechanism, and determine the conditions under which the observed energy gain emerges and subsequently breaks down. The second (methodological) objective is to *validate the open-source propagator* used for the first: to cross-check its force models, component by component, against the ~~high-fidelity~~ industrial flight-dynamics tool STK Astrogator and against the independent open-source library Orekit, and to quantify where the ~~two~~ tools agree and diverge. The two objectives are complementary — the validation establishes that the gain region is reproducible across independent tools and identifies which force-model choices most strongly govern the result.

Several exclusions are imposed to maintain a controlled scope. This study does not claim disposal-compliant mission design, communications-feasible spacecraft architecture, collision-risk-assessed operations, maintenance of an exact Sun-synchronous orbit, or long-term sustained orbit raising. It does not attempt subsystem-level mission design, active attitude-control implementation, or guidance and feedback control. The external validation performed here is a component-level cross-check of the propagator's force models against STK Astrogator and Orekit, not a flight-data validation of the mission concept; the results should therefore not be interpreted as a demonstrated mission concept, but as evidence for a bounded dynamical effect within the modelled regime, together with a verified toolchain for studying it. Sun-synchronous-like initial conditions are used only to impose a controlled and interpretable Sun-relative geometry; they are not intended to assert that the vehicle remains on an exactly maintained operational SSO throughout the propagation.

3. Physical Mechanism of Eclipse-Modulated Energy Pumping

This section introduces a reduced analytical picture to isolate the core geometric mechanism by which eclipse timing can bias solar-radiation-pressure (SRP) work over an orbit. Its purpose is not to derive a design-optimal sail orientation for the full propagated problem, nor to replace the later

3D simulations, but to establish the sign logic of phase-selective forcing in the simplest setting where it can be seen clearly. The full numerical problem considered later includes three-dimensional orbital geometry, J_2-driven nodal precession, drag, eclipse-modulated illumination, and coupled evolution of the orbital elements; the reduced model is therefore used here only as a mechanistic scaffold for interpreting those results.
The reduced argument developed below is used to identify the sign structure of the SRP work integral under eclipse-modulated illumination, and thereby to motivate the later propagated results.

3.1. ***Orbital energy change and transverse work***

The central quantity of interest is the sign of the orbit-averaged transverse SRP work, because the Gauss variational equation for semi-major axis [32, 33] directly links da/dt to the radial and transverse components of the perturbing acceleration. Writing the perturbing acceleration in the local orbital frame as radial (F_R), transverse (F_T), and normal (F_N) components, the semi-major-axis equation is

$$\frac{da}{dt} = \frac{2a^2}{h}\left[e\sin\theta\, F_R + (1 + e\cos\theta)\, F_T\right], \tag{1}$$

where F_R and F_T are the radial and transverse components of the perturbing acceleration, θ is the true anomaly, a is the semi-major axis, and h is the specific angular momentum.

For near-circular low Earth orbits ($e \approx 0$), this reduces to

$$\frac{da}{dt} \approx \frac{2a^2}{h} F_T = \frac{2}{n} F_T, \tag{2}$$

where $n = \sqrt{\mu/a^3}$ is the mean motion. The semi-major axis evolution is therefore governed primarily by the transverse component of the perturbing force. Equivalently, since the specific orbital energy is

$$\varepsilon = -\frac{\mu}{2a}, \tag{3}$$

its rate of change is

$$\dot{\varepsilon} = \mathbf{a}_p \cdot \mathbf{v} = F_T v, \tag{4}$$

for a circular orbit, where $\mathbf{a}_p$ is the perturbing acceleration and $v = \sqrt{\mu/a}$ is the orbital speed. Hence, in the near-circular setting used here, positive

transverse work corresponds to increasing orbital energy and increasing semi-major axis, while negative transverse work has the opposite effect.

This makes the sign of the orbit-averaged transverse work the central quantity of interest in the reduced mechanism. In particular, if the SRP contribution satisfies

$$\int_0^T F_T \, dt > 0, \tag{5}$$

over one orbital period T, then the orbit-integrated SRP work is positive and the corresponding one-orbit change in orbital energy, and hence in semi-major axis, is favourable.

3.2. ***Passive symmetry and cancellation in the fully illuminated case***

Consider first an idealised 2D planar circular orbit with the Sun fixed in inertial space. Let the Sun direction define the $+x$ axis, and parameterise the spacecraft position and velocity by

$$\mathbf{r} = r(\cos\theta, \sin\theta), \qquad \mathbf{v} = v(-\sin\theta, \cos\theta), \tag{6}$$

where θ is the orbital phase measured from the sunward direction. For a passively oriented, spin-stabilised sail with fixed inertial attitude, let the sail normal be

$$\mathbf{n} = (\cos\beta, \sin\beta), \tag{7}$$

where β is the constant pitch angle measured from the Sun line.

For an ideal reflecting sail under constant illumination, ~~the SRP acceleration~~ with $\mathbf{n}$ the normal of the illuminated (sunward) face, the SRP acceleration is directed away from the Sun along $-\mathbf{n}$, has fixed inertial direction, and may be written in the form [14]

$$\mathbf{a}_{\mathrm{SRP}} = -a_0 \cos^2\beta \, \mathbf{n}, \tag{8}$$

where a_0 is the characteristic SRP acceleration scale. Equation (8) is the ideal-reflector limit of the flat-plate force used in the simulations (Eq. (30) in Section 4): a real plate with specular reflectivity $\rho_s < 1$ also carries an absorbed component directed along the Sun line, but that component performs no net work over an orbit in this geometry, so only

the normal-directed term enters the energy argument that follows. The instantaneous specific power is then

$$\dot{\varepsilon} = \mathbf{a}_{\mathrm{SRP}} \cdot \mathbf{v} = -a_0 \cos^2 \beta \, \mathbf{n} \cdot \mathbf{v}. \tag{9}$$

Substituting for $\mathbf{n}$ and $\mathbf{v}$ gives

$$\mathbf{n} \cdot \mathbf{v} = v(-\sin\theta\cos\beta + \cos\theta\sin\beta) = v\sin(\beta - \theta), \tag{10}$$

so that

$$\dot{\varepsilon} = -a_0 v \cos^2 \beta \, \sin(\beta - \theta). \tag{11}$$

Over a fully illuminated circular orbit, using $dt = (r/v)\, d\theta$, the net specific energy change is

$$\Delta\varepsilon_{\mathrm{full}} = \int_0^{2\pi} \dot{\varepsilon}\, dt = -a_0 r \cos^2 \beta \int_0^{2\pi} \sin(\beta - \theta)\, d\theta. \tag{12}$$

Since

$$\int_0^{2\pi} \sin(\beta - \theta)\, d\theta = 0, \tag{13}$$

it follows that

$$\Delta\varepsilon_{\mathrm{full}} = 0. \tag{14}$$

Equation (14) formalises the passive symmetry argument: under continuous illumination, a fixed-attitude sail experiences alternating orbital phases of positive and negative transverse work that cancel exactly over one revolution. In this idealised baseline case, the one-orbit net change in orbital energy vanishes despite non-zero instantaneous SRP forcing.

3.3. ***Eclipse as a selective gating mechanism***

The previous cancellation result changes fundamentally once eclipse is introduced. In a midnight orbit under a cylindrical shadow model, the eclipse is centred on the anti-solar direction $\theta = \pi$ and spans the interval

$$\theta \in [\pi - \delta, \pi + \delta], \qquad \delta = \arcsin\left(\frac{R_E}{r}\right), \tag{15}$$

where R_E is the Earth radius and r is the orbital radius. The illuminated arcs are therefore

$$[0, \pi - \delta] \cup [\pi + \delta, 2\pi]. \tag{16}$$

The net specific energy change over the illuminated portion of the orbit becomes

$$\Delta\varepsilon_{\mathrm{lit}} = -a_0 r \cos^2\beta \left[\int_0^{\pi-\delta} \sin(\beta-\theta)\, d\theta + \int_{\pi+\delta}^{2\pi} \sin(\beta-\theta)\, d\theta \right]. \tag{17}$$

Using the full-orbit cancellation result, this may be written more compactly as minus the contribution from the shadowed arc,

$$\Delta\varepsilon_{\mathrm{lit}} = a_0 r \cos^2\beta \int_{\pi-\delta}^{\pi+\delta} \sin(\beta-\theta)\, d\theta. \tag{18}$$

Evaluating the integral gives

$$\int_{\pi-\delta}^{\pi+\delta} \sin(\beta-\theta)\, d\theta = \Big[\cos(\beta-\theta)\Big]_{\pi-\delta}^{\pi+\delta} = -2\sin\beta\,\sin\delta, \tag{19}$$

and hence

$$\Delta\varepsilon_{\mathrm{lit}} = -2a_0 r \cos^2\beta\,\sin\beta\,\sin\delta. \tag{20}$$

Equation (20) shows explicitly that eclipse does not merely reduce the SRP magnitude; it breaks the passive cancellation by removing a finite interval of orbital phase from the work integral. The sign of the net energy change is therefore controlled by the sign of $-\sin\beta$. In this 2D midnight-orbit idealisation,

$$\Delta\varepsilon_{\mathrm{lit}} > 0 \qquad \Longleftrightarrow \qquad \sin\beta < 0. \tag{21}$$

A finite branch of pitch angles therefore exists for which eclipse suppresses unfavourable work more strongly than favourable work, leaving a positive residual contribution to the one-orbit SRP work balance and hence a positive one-orbit energy change.

Within this reduced 2D midnight-orbit model, the magnitude of the one-orbit energy gain is maximised by maximising $-\sin\beta\cos^2\beta$, which gives

$$\sin^2\beta = \frac{1}{3}, \tag{22}$$

with the positive-gain branch

$$\beta^\star = -\arcsin\left(\frac{1}{\sqrt{3}}\right) \approx -35.3^\circ. \tag{23}$$

In the signed-pitch convention of Section 4 ($p = -\beta$) this branch is $p^\star \approx +35°$, which is where the propagated 800 km map of Section 6 peaks. This result is used here only to show that the eclipse-modulated work imbalance is not marginal or accidental in the reduced geometry. It is not claimed to be a design optimum for the full propagated 3D system, whose geometry, illumination history, drag losses, and coupled element evolution are substantially richer.

3.4. ***Dependence on orbital geometry***

The reduced mechanism is intrinsically phase-selective and therefore strongly dependent on orbital geometry. In the full problem, the local time of ~~descending node (LTDN~~ascending node (LTAN), the Sun-relative orientation of the orbital plane, and the fixed sail attitude together determine which parts of the orbit are illuminated, which are shadowed, and how the SRP work distribution is sampled. The sign and magnitude of the orbit-integrated SRP contribution therefore depend on whether eclipse preferentially removes unfavourable or favourable work phases.

The midnight case is analytically useful because the eclipse is centred on the anti-solar point and the resulting shadow interval is symmetric in orbital phase, allowing the effect of selective gating to be shown in closed form. Away from this limiting geometry, the illuminated and shadowed arcs shift relative to the SRP work distribution, and the balance between positive and negative transverse work changes accordingly. Net semi-major-axis growth is therefore expected only over restricted regions of the full parameter space.

Crucially, the same phase-dependent forcing that produces a favourable one-orbit SRP work imbalance need not produce shape-preserving orbit raising. Gauss' variational equations imply that the detailed phasing of F_R and F_T can simultaneously drive changes in eccentricity and related orbital geometry, while out-of-plane forcing in the full 3D problem affects inclination and node. A positive orbit-averaged SRP work contribution is therefore not equivalent to uniform upward translation of the orbit. In the propagated cases examined later, the same eclipse-biased phasing that produces transient semi-major-axis gain can coexist with eccentricity growth, altered perigee and apogee behaviour, and ultimately degradation of the gain regime.

3.5. Transient nature of the mechanism

The reduced analysis establishes only that eclipse can bias the one-orbit SRP work balance in favour of net energy gain for restricted geometries. It does not establish persistence, monotonic orbit raising, or long-horizon robustness once the full coupled dynamics are restored. The propagated 3D results therefore test whether this favourable work asymmetry survives in realistic near-polar LEO motion, and how it breaks down through coupled changes in eccentricity, drag exposure, and Sun-relative geometry; these longer-horizon effects are examined in Section 7.

4. Dynamical model and assumptions

This study considers a bounded orbital-dynamics problem in near-polar low Earth orbit (LEO) intended to isolate the short-horizon interaction between solar radiation pressure (SRP), eclipse timing, atmospheric drag, and J_2-driven nodal precession. The emphasis is on whether a sail with inertially fixed orientation can exhibit short-horizon semi-major-axis growth while remaining close to Sun-synchronous orbital geometry.

A Sun-synchronous orbit (SSO) is desirable because it maintains a predictable Sun-relative geometry while allowing the orbital plane to precess with the apparent annual motion of the Sun [32]. For near-circular LEO, this requires a secular right ascension of the ascending node (RAAN) precession rate of approximately

$$\frac{d\Omega}{dt} \approx 0.986^\circ/\text{day}, \tag{24}$$

which is primarily supplied by Earth's oblateness through the J_2 perturbation. The corresponding nodal precession rate is [32]

$$\dot{\Omega} = -\frac{3}{2} J_2 \left(\frac{R_E}{a(1-e^2)} \right)^2 \sqrt{\frac{\mu}{a^3}} \cos i. \tag{25}$$

For circular orbits ($e = 0$) in the altitude range considered here, the inclination required to satisfy Eq. (24) is close to 98°, as summarised in Table 1. The simulations use $i_0 = 98.17^\circ$ at 700 km and 98.31° at 800 km (Table 2). The former is within 0.02° of the Sun-synchronous value; the latter, inherited from the original study design, lies 0.29° below it, so that

at 800 km the node precesses 0.033° per day more slowly than the mean Sun and the local time of the node lags by about 10 minutes over the 75-day propagations. This departure is immaterial on the 14-day horizon of the parameter maps and small compared with the Sun–normal drift that governs the long-horizon behaviour (Section 7); the orbits are therefore described as near-Sun-synchronous throughout.

Table 1: Inclination required for a Sun-synchronous circular orbit ($e = 0$) at selected altitudes, from Eq. (24) with $J_2 = 1.0826 \times 10^{-3}$, $R_E = 6378.137$ km and a nodal rate of 0.9856° per day.

Altitude (km)	**Required Inclination (°)**
600	97.79
700	98.19
800	98.60
900	99.03

Table 2: Baseline parameters used for the representative gain-case propagation and local mechanism analysis. The representative geometries (LTAN, pitch) are listed in Table 5.

Parameter	**Value**
Epoch (initial state, Sun geometry)	1 Jan 2026, 12:00:00 UTC
Initial altitude, h_0 (production / contrast)	800 km / 700 km
Initial inclination, i_0 (near-Sun-synchronous, see Table 1)	98.31° / 98.17°
Initial eccentricity, e_0; ω_0; ν_0	1×10^{-4}; 0°; 0°
Spacecraft mass, m	10 kg
Sail area, A (one-sided 10×10 m plate)	100 m^2
Area-to-mass ratio, A/m	10 m^2 kg^{-1}
Optical coefficients $(\alpha_a, \rho_s, \rho_d)$; $C_r = 1 + \rho_s$	$(0.3, 0.7, 0)$; 1.7
Drag coefficient, C_D	2.2
Integrator; step (maps and cases / one-orbit diagnostics)	RK4; 10 s / 2 s

In the present work, the sail orientation is prescribed as inertially fixed over the time horizons of interest, and no active steering law is applied during propagation. This contrasts with actively steered sails [34, 35] and with active momentum management approaches such as those developed for Solar

Cruiser [36, 37]. The fixed-attitude assumption is deliberate: it allows the orbital consequences of a fixed force geometry to be isolated directly, without introducing control-law dependence into the interpretation of the results. At the same time, it imposes an important limitation. As the orbital plane precesses and the Sun-relative geometry evolves, the orientation of the sail with respect to both the Sun and the local orbital frame cannot remain optimal indefinitely. Any favourable energy-transfer regime must therefore be understood as geometry-bounded and transient rather than as a permanently sustained operating mode.

The evolution of the orbit under weak non-Keplerian perturbations is interpreted using the Gauss Variational Equations (GVEs) [32, 33]. Writing the perturbing acceleration vector $\vec{F}$ in the local orbital frame as radial (F_R), transverse (F_T), and normal (F_N) components, the orbital-element evolution is given by

$$\frac{da}{dt} = \frac{2a^2}{h}\left[e\sin\theta\, F_R + (1 + e\cos\theta)\, F_T\right], \tag{26}$$

$$\frac{de}{dt} = \frac{h}{\mu}\left[\sin\theta\, F_R + \left(\cos\theta + \frac{e + \cos\theta}{1 + e\cos\theta}\right) F_T\right], \tag{27}$$

$$\frac{di}{dt} = \frac{r\cos u}{h}\, F_N, \tag{28}$$

$$\frac{d\Omega}{dt} = \frac{r\sin u}{h\sin i}\, F_N, \tag{29}$$

where $h = \sqrt{\mu a(1 - e^2)}$ is the specific angular momentum, $r = a(1-e^2)/(1+e\cos\theta)$ is the instantaneous orbital radius, θ is the true anomaly, and $u = \omega + \theta$ is the argument of latitude. A detailed derivation of the GVEs is given by Vallado [32].

These expressions clarify the role of each perturbing component. In near-circular LEO, semi-major-axis growth is driven primarily by the in-plane force, especially the transverse component F_T, whereas inclination i and RAAN Ω evolve through the normal component F_N. Since Sun-synchronicity relies on the secular J_2-driven nodal precession described above, additional out-of-plane forcing should remain small if the orbit is to stay close to the desired SSO geometry.

This creates the central geometrical tension of the present study. To obtain semi-major-axis growth, the sail must produce a favourable in-plane force

with a strong transverse projection, while remaining close to SSO requires the normal component F_N to stay limited. Because the sail orientation is fixed in inertial space, the Sun-relative cone angle and the orbital-frame force decomposition drift as the orbit evolves, so any favourable geometry is expected to persist only over a bounded interval.

The force model retained for this study is intentionally simple and is chosen to isolate the mechanism of interest rather than to maximise environmental fidelity. The translational dynamics include central gravity, Earth's oblateness through the J_2 term, solar radiation pressure, eclipse shadowing, and atmospheric drag. No Earth albedo model is included.

Solar radiation pressure

The SRP model is based on photon momentum transfer to a reflective flat plate [14]. For an idealised square sail with cone angle α relative to the Sun line, the force magnitude follows the familiar $\cos^2\alpha$ dependence for a flat-plate sail (see 14, Ch. 2). In the implementation used here, the physical flat-plate decomposition into absorbed, specularly reflected and diffusely reflected components [14, 38, 39], i.e. the same force law as the N-plate models of flight-dynamics tools. Let $\hat{\mathbf{s}}$ be the unit vector from the Sun to the spacecraft, $\hat{\mathbf{n}}$ the sail normal (the sail is illuminated on one side only, so SRP acts only when $\cos\alpha = -\hat{\mathbf{n}}\cdot\hat{\mathbf{s}} > 0$, where α is the incidence angle between the normal and the Sun line), and $\alpha_a + \rho_s + \rho_d = 1$ the absorptivity, specular reflectivity and diffuse reflectivity. The SRP acceleration is written as

$$\mathbf{a}_{\mathrm{SRP}} = \frac{P_{1\ \mathrm{AU}}\, A}{m}\left(\frac{1\ \mathrm{AU}}{d_\odot}\right)^2 \left[(\alpha_a + \rho_d)\cos\alpha\ \hat{\mathbf{s}} \;-\; \left(2\rho_s\cos^2\alpha + \tfrac{2}{3}\rho_d\cos\alpha\right)\hat{\mathbf{n}}\right], \tag{30}$$

where $P_{1\ \mathrm{AU}}$ denotes $P_{1\ \mathrm{AU}} = 4.56\times10^{-6}\,\mathrm{N\,m^{-2}}$ is the solar radiation flux pressure at one astronomical unit [40], (a total solar irradiance of $1367\,\mathrm{W\,m^{-2}}$ divided by the speed of light, the value conventionally used in sail studies [14]; the modern value of $1361\,\mathrm{W\,m^{-2}}$ [40] is 0.4% lower, which is immaterial here), A is the sail area, C_r is the reflectivity coefficient, m is the spacecraft mass , and $\hat{\mathbf{r}}$ is the unit vector from the spacecraft to the Sun. The SRP and $d_\odot$ the instantaneous Earth–Sun distance from a low-precision analytic ephemeris (the epoch lies within days of perihelion, so the flux exceeds its 1 AU value by up to 3.4%). The optical coefficients are $\rho_s = 0.7$, $\rho_d = 0$, $\alpha_a = 0.3$, giving the normal-incidence coefficient $C_r = \alpha_a + 2\rho_s + \frac{5}{3}\rho_d = 1 + \rho_s = 1.7$ of Table 2. The specular term acts along the plate normal with the $\cos^2\alpha$ law of

Eq. (8), and it is this along-normal component that carries the eclipse-gated mechanism of Section 3; the smaller absorbed term acts along the Sun line and, like any force of fixed inertial direction along $\hat{\mathbf{s}}$, performs no net work over an orbit whose shadow arc is symmetric about the anti-solar point (Section 9). The acceleration is evaluated in the inertial frame and then resolved into the orbital frame for interpretation through the GVEs.

Eclipses are modelled using a binary ~~conical~~ cylindrical shadow geometry. The spacecraft is treated as illuminated when it lies outside the Earth~~'s umbral cone~~'s shadow cylinder (the umbra, with the Sun taken at infinity) and fully shadowed when it lies within it, so that SRP is either applied in full or suppressed entirely; no penumbra is modelled. This treatment is sufficient for the present purpose because the paper is concerned primarily with the orbital-phase selectivity of SRP application rather than with resolving penumbral transition details. The shadowed SRP acceleration is therefore written as

$$\mathbf{a}_{\mathrm{SRP}}^{\mathrm{shadowed}} = s \cdot \mathbf{a}_{\mathrm{SRP}}, \tag{31}$$

where

$$s = \begin{cases} 1, & \text{illuminated,} \\ 0, & \text{eclipsed.} \end{cases}$$

This eclipse treatment is central to the present paper because the mechanism of interest depends not only on the magnitude of SRP, but on which orbital arcs remain illuminated and which are removed by shadow.

Attitude and node geometry

The sail normal is constructed once, at epoch, and then held fixed in inertial space. Let $\hat{\mathbf{s}}_\odot = -\hat{\mathbf{s}}$ be the unit vector from the spacecraft towards the Sun and $\hat{\mathbf{h}}$ the orbit normal at epoch; the in-plane direction perpendicular to the Sun line is $\hat{\mathbf{b}} = \hat{\mathbf{s}}_\odot \times \hat{\mathbf{h}} / |\hat{\mathbf{s}}_\odot \times \hat{\mathbf{h}}|$, and the normal of the illuminated face is

$$\hat{\mathbf{n}} = \cos p\, \hat{\mathbf{s}}_\odot + \sin p\, \hat{\mathbf{b}}, \tag{32}$$

rotated, where a roll angle is prescribed (Section 8), about $\hat{\mathbf{s}}_\odot$. The pitch p is therefore signed: positive pitch tilts the illuminated face from the Sun line towards $\hat{\mathbf{b}}$, negative pitch towards $-\hat{\mathbf{b}}$, and $|p| \to 90°$ turns the plate edge-on to the Sun; in the two-dimensional picture of Section 3, where the orbit is traversed anticlockwise about $\hat{\mathbf{h}}$, the pitch of Eq. (32) corresponds

to $\beta = -p$. The orbital plane is specified by the local time of the ascending node, which fixes the right ascension of the node at epoch through

$$\Omega = \alpha_\odot + 15^\circ\,(\mathrm{LTAN} - 12\,\mathrm{h}), \tag{33}$$

with $\alpha_\odot = 282.05^\circ$ the right ascension of the Sun at the epoch of Table 2; LTAN $= 12\,\mathrm{h}$ is the noon–midnight plane and LTAN $= 6\,\mathrm{h}$ or $18\,\mathrm{h}$ the dawn–dusk plane.

Atmospheric drag

Atmospheric drag is included because it remains dynamically relevant throughout the altitude range considered, particularly once eccentricity growth lowers perigee and increases local density exposure. The atmospheric density is ~~represented using a piecewise-exponential fit to~~ taken from the U.S. Standard Atmosphere ~~(~~1976 ~~) data [41] based on Braeunig 's tabulation [42]~~ [41] as tabulated by Braeunig [42] and interpolated exponentially between the tabulated altitudes (every 100 km above 500 km). Within each ~~altitude~~ interval i ~~,~~ the density is evaluated as

$$\rho(h) = \rho_i \exp\left(-\frac{h - h_i}{H_i}\right), \qquad H_i = \frac{h_{i+1} - h_i}{\ln\left(\rho_i/\rho_{i+1}\right)}, \tag{34}$$

~~where $\rho_0^{(i)}$, $h_0^{(i)}$, and $H^{(i)}$ are the reference density, reference altitude, and scale height for that layer~~so that the profile is continuous and passes through every tabulated value ($3.070 \times 10^{-14}\,\mathrm{kg\,m^{-3}}$ at 700 km, $1.136 \times 10^{-14}\,\mathrm{kg\,m^{-3}}$ at 800 km). The density is evaluated at the geocentric altitude. The profile is static: it carries no dependence on solar flux or geomagnetic activity, a limitation whose consequences are quantified in Section 9.

The aerodynamic model used here is a simple projected-area drag law. Atmospheric rotational velocity is not modelled; the relative flow is taken directly from the spacecraft inertial velocity. For the near-polar orbits considered here this is a small convention: switching on a rigidly co-rotating atmosphere changes the 14-day drag-induced decay by +3.4% at 700 km and +2.4% at 800 km, and evaluating the density at geodetic rather than geocentric altitude changes it by −18% and −10% (both quantified in Section 9); the choice of atmosphere *model* matters far more. The drag acceleration is written as

$$\mathbf{a}_{\mathrm{drag}} = -\frac{1}{2}\,\rho(h)\,v^2\,C_D\,\frac{A_{\mathrm{frontal}}}{m}\,\hat{\mathbf{v}}, \tag{35}$$

where $C_D = 2.2$ is a standard value for satellites in free-molecular flow [32], m is the spacecraft mass, $v = \|\mathbf{v}\|$, and $\hat{\mathbf{v}}$ is the unit vector in the velocity direction. The frontal area A_{frontal} is computed from the projection of the sail surface onto the plane normal to the instantaneous velocity vector, so that the effective drag area varies with the angle between the inertially fixed sail normal and the flow.

The drag model is intentionally simple. Its role in this paper is not to resolve high-fidelity free-molecular aerodynamics, but to capture the dominant loss mechanism opposing SRP-driven energy gain. This is especially important once the orbit departs from circularity, because drag then remains concentrated near perigee and can overtake the orbit-averaged SRP surplus even when early-time semi-major-axis growth is observed.

The assumptions adopted above are sufficient for the present purpose of mechanism isolation and regime mapping. They do not constitute a full validation of passive sail performance over long durations, nor do they resolve all structural, control, or environmental uncertainties relevant to flight operations. Those questions are outside the scope of this section and are addressed only insofar as they affect interpretation of the short-horizon transient dynamics studied in this paper.

Table 3: Summary of the principal modelling assumptions used in the orbital propagations.

Assumption / model element	**Implementation used in this study**
Attitude treatment	Sail orientation prescribed as inertially fixed over the horizon of interest; no active steering law applied during propagation.
Central gravity	Included as the baseline point-mass Earth gravitational field.
J_2 perturbation	Included to capture Earth oblateness and the associated secular nodal precession required for near-Sun-synchronous behaviour.
Solar radiation pressure (SRP)	Included using a flat-plate sail model with one-sided illumination and force scaling with incidence angle through a cosine-based projection law.
Eclipse model	Binary cylindrical Earth-shadow (umbra) model that suppresses SRP entirely while the spacecraft is in shadow; no penumbra.
Atmospheric drag	Included using a projected-area flat-plate drag law with $C_D = 2.2$.
Density model	U.S. Standard Atmosphere 1976, exponential interpolation between the tabulated values; static (no solar-activity dependence).
Atmospheric co-rotation	Neglected; relative flow taken directly from the spacecraft inertial velocity.
Earth albedo	Omitted.
Orbit regime	Near-circular, near-polar LEO focused on maintaining approximate Sun-synchronous geometry.
Study objective	Controlled mechanism-isolation study of transient gain behaviour, not a full mission-feasibility or flight-operations model.

5. Parameter Space and Metrics

Having defined the governing assumptions and force models, we now specify the parameter space and, critically, the fixed-window metric used to identify the transient gain regime. Because the central question of this paper is whether eclipse-modulated SRP can produce a *temporary* semi-major-axis increase over a bounded horizon, the primary map-based quantity is not the final change in a at the end of the run, and not the long-horizon change over 75 days, but the largest positive excursion above the initial semi-major axis attained at any time within a prescribed analysis window. The aim is therefore to determine whether such transient gain occupies a coherent region of design space, and how narrow or robust that region is.

The primary parameters varied in this study are the local time of the ~~descending node (LTDN~~ascending node (LTAN), the initial sail orientation, and the initial orbital altitude. ~~LTDN~~ LTAN determines the Sun-relative geometry of the orbit and hence the phasing between illuminated arcs, eclipsed arcs, and the SRP work pattern; the initial sail orientation controls the decomposition of SRP into radial, transverse, and normal components; and altitude alters both the drag environment and the baseline Sun-synchronous inclination required through ~~24~~Eq. (24). Together, these parameters define the geometry within which the transient gain regime may or may not appear.

In the baseline sweeps, the initial orbit is taken to be near-circular and ~~Sun-synchronous, with inclination selected consistently with~~ near-Sun-synchronous, with the inclination of Table 2 for the chosen altitude ~~using Table 1~~(Section 4). For each case, the sail orientation is prescribed as inertially fixed from the initial epoch onward. Additional sensitivity studies may then be introduced around representative gain and non-gain cases by perturbing the nominal sail orientation, varying atmospheric density through a scalar multiplier, or applying small static orientation offsets to test the fragility of the identified regime.

The parameter-space exploration is intentionally staged. The first sweep is performed over ~~LTDN~~ LTAN and initial sail angle at fixed altitude and fixed sail loading, in order to determine whether a connected region of *transient* gain exists at all under the fixed-window metric defined below. Only after this primary map is established are additional altitude cases and uncertainty sweeps introduced.

Table 4 lists the parameter ranges used. The LTAN grid stops at 23.5 h

because 24 h is the same orbital plane as 0 h; the map is periodic in LTAN and is read accordingly. Pitch is signed: positive pitch tilts the sail normal from the Sun line towards the in-plane direction $\hat{\mathbf{s}}_\odot \times \hat{\mathbf{h}}$ (Section 4); negative pitch tilts it the other way. Beyond $\pm 90°$ the one-sided sail is not illuminated, so that range is not sampled.

Table 4: Parameter ranges of the production sweeps and sensitivity studies. All runs use the baseline spacecraft of Table 2 unless a row states otherwise.

Study	**Parameter**	**Range / values**
LTAN–pitch map	altitude	800 km (production), 700 km (contrast)
	LTAN	0 to 23.5 h, step 0.5 h (48 values)
	pitch	$-90°$ to $+90°$, step $5°$ (37 values)
	horizon; metric	14 d; $\Delta a_{\max,14\mathrm{d}}$ (and $\Delta a(14\,\mathrm{d})$)
	cells per altitude	1776
Representative cases	horizon	14 d (eclipse on/off), 75 d (breakdown)
Roll sensitivity	roll about the Sun line	$-45°$ to $+45°$, 19 values
Density sensitivity	multiplier on $\rho(h)$	$0.25, 0.5, 1, 2, 4$
Area-to-mass sensitivity	A/m at fixed $m = 10\,\mathrm{kg}$	2.5 to 22.5 $\mathrm{m^2\,kg^{-1}}$, 9 values
Step-size convergence	RK4 step	$120, 60, 30, 15, 10, 7.5, 5, 2.5$ s

A fixed analysis horizon is used for consistent comparison between cases. The primary map-based metric is

$$\Delta a_{\max}(T) = \max_{t\in[0,T]} \left(a(t) - a_0\right), \tag{36}$$

with $T = 14\,\mathrm{d}$ in the principal parameter-space sweep. Here $a(t)$ is the mean (secular) semi-major axis, obtained by sampling the osculating value at successive ascending-node crossings so that the ± 9 km J_2 short-period oscillation is removed; a_0 is the value at the first such crossing, one nodal period after epoch, so every Δa quoted in this paper excludes the secular change of the first orbit (below 0.1 km in all cases). Thus, $\Delta a_{\max,14\mathrm{d}}$ measures the largest positive semi-major-axis excursion above the initial orbit attained *anywhere* within the first 14 days; it is zero when the mean semi-major

axis never rises above a_0. It is *not* the final value of $a(14\,\mathrm{d}) - a_0$, and it is *not* the net change after the longer 75-day propagation used later to study breakdown. This choice is deliberate: for a transient-gain problem, the physically relevant first question is whether a trajectory ever enters a favourable interval of positive semi-major-axis excursion within a common bounded window, not whether that excursion survives indefinitely.

To support physical interpretation, the fixed-window excursion metric is supplemented by orbit-resolved work diagnostics. In particular, we track the cumulative contribution of SRP and drag to the along-track work budget, since the mechanism proposed in Section 3 is not merely that SRP is present, but that eclipse can selectively retain or suppress SRP work over specific orbital arcs and thereby produce, or fail to produce, a temporary positive excursion in a. For a given perturbing acceleration $\mathbf{a}_p$, the instantaneous specific power is

$$P_p(t) = \mathbf{a}_p(t) \cdot \mathbf{v}(t), \tag{37}$$

and the associated cumulative specific work over a time interval $[t_1, t_2]$ is

$$W_p(t_1, t_2) = \int_{t_1}^{t_2} \mathbf{a}_p(t) \cdot \mathbf{v}(t)\, dt. \tag{38}$$

In practice, these quantities are evaluated separately for SRP and drag and then accumulated over individual orbits and over the full analysis horizon.

Because the hypothesised mechanism is most naturally interpreted in the orbital frame, we also track the transverse components of the perturbing accelerations. From Eq. (26), the transverse contribution is especially important in near-circular orbits, where it acts most directly on orbital energy and semi-major axis. For this reason, we record both the instantaneous transverse accelerations $F_{T,\mathrm{SRP}}$ and $F_{T,\mathrm{drag}}$, and their orbit-integrated contributions, alongside the full energy-based work budget. These diagnostics are used to distinguish between three physically different situations: cases in which eclipse removes predominantly unfavourable SRP arcs, cases in which eclipse produces little net bias, and cases in which the remaining illuminated arcs are themselves energetically unfavourable.

In addition to semi-major axis, we record the evolution of apogee radius r_a, perigee radius r_p, eccentricity e, inclination i, and right ascension of the ascending node Ω. These quantities are necessary because positive Δa alone does not imply clean whole-orbit raising. In particular, a gain in a may be

accompanied by growth in eccentricity, with apogee increasing more strongly than perigee. Since drag remains concentrated near perigee, this distinction is central to understanding why the transient gain regime can disappear even after an initially favourable interval. The coupled evolution of r_a, r_p, and e therefore forms a second layer of interpretation beyond the headline Δa result.

To capture the role of eclipse geometry explicitly, we also record the eclipse state as a binary quantity throughout propagation, together with the eclipse fraction of each orbit. This allows the work budget to be partitioned orbit-by-orbit into illuminated and shadowed segments. In representative cases, these diagnostics are later plotted against true anomaly so that the reader can see directly which orbital arcs contribute positive SRP work, which contribute negative SRP work, and which are removed by eclipse. This is an important distinction: the paper does not argue merely that eclipse reduces SRP, but that eclipse can alter the sign balance of orbit-averaged SRP work.

On the basis of the fixed-horizon metric $\Delta a_{\max}(T)$, each simulation is assigned to one of three qualitative classes:

- **Gain case:** $\Delta a_{\max}(T) > 0$, indicating that the trajectory attains a positive semi-major-axis excursion at some point within the window;
- **Neutral case:** $\Delta a_{\max}(T) \approx 0$, indicating that no meaningful positive excursion is attained within the window;
- **Decay case:** cases for which $a(t) - a_0$ remains non-positive throughout the window, so that $\Delta a_{\max}(T)$ is zero or numerically negligible and the trajectory is effectively monotone non-gaining over that interval.

These labels are descriptive rather than ontological. In particular, "gain" here means *transient gain within the prescribed window*, not sustained long-horizon orbit raising; likewise, the strongest gain cell in the map is the case with the largest $\Delta a_{\max,14\mathrm{d}}$, not necessarily the most favourable long-horizon trajectory once eccentricity growth and drag accumulation are taken into account.

For representative cases drawn from each class, we additionally identify a transient gain window from the cumulative SRP and drag work curves. This secondary diagnostic is used later to distinguish early favourable behaviour

from the subsequent drag-dominated breakdown phase. It is therefore interpretive rather than classificatory: the parameter-space map itself is defined only by $\Delta a_{\max,14\mathrm{d}}$.

The minimum set of diagnostics recorded for each run is therefore as follows: instantaneous SRP and drag accelerations, their transverse components, eclipse state, classical orbital elements, cumulative SRP and drag work, orbit-by-orbit eclipse fraction, and the evolution of a, r_a, and r_p. This choice is deliberate. The present paper is a mechanism and regime-mapping study, so the recorded metrics must do more than rank cases numerically: they must also explain why a given case gains, remains neutral, or decays.

The results that follow therefore proceed in three steps. First, the fixed-window metric $\Delta a_{\max,14\mathrm{d}}$ is used to identify where transient gain does and does not occur in parameter space. Second, orbit-resolved work budgets are used to test whether eclipse causally biases the SRP work balance in those cases. Third, apogee, perigee, eccentricity, and cumulative drag/SRP comparisons are used to explain why a case that exhibits transient gain over 14 days need not remain favourable over longer horizons.

6. Existence and Mechanism of the Transient Gain Regime

This section addresses four questions in sequence: whether a bounded ~~transient-gain region exists at 700 km,~~ region of transient gain exists in LTAN–pitch space, and at which of the two altitudes of Table 2; whether eclipse is causally responsible for it~~,~~; how that causality appears in the orbit-resolved SRP work budget~~,~~; and why the representative gain case should ~~still~~ be interpreted as transient rather than sustained. ~~Because the paper is concerned with *transient* gain, the~~ The map-based metric ~~used here is the maximum semi-major-axis increase attained within the first~~ is $\Delta a_{\max,14\mathrm{d}}$ of Section 5, the largest positive excursion of the mean (node-sampled) semi-major axis above a_0 within 14 days, ~~not the final Δa at the end of the run~~zero if there is none; the end-of-window value $\Delta a(14\,\mathrm{d})$ is quoted alongside it.

Figure 2 presents the ~~fixed-window map of maximum semi-major-axis increase at~~ 14-day maps of $\Delta a_{\max,14\mathrm{d}}$ at 800 km (left) and 700 km ~~as a function of LTDN~~ (right) over the 48×37 grid of LTAN (0 to 23.5 h, step 0.5 h) and initial pitch ~~angle. A single dominant positive lobe appears over approximately LTDN 9.0~~ ($-90°$ to ~~14.75 h and pitch 0°~~ $+90°$, step 5°),

1776 cells per altitude. At 800 km, 346 of the 1776 cells (19.5%) show a positive excursion within 14 days; 255 exceed 1 km, 155 exceed 3 km and 47 exceed 5 km. The peak is $\Delta a_{\max,14\mathrm{d}} = 5.95$ km at LTAN 14.0 h, pitch +35° (star). Gain occurs only at positive pitch, between +10° and +60°, with the ridge at 35° to ~~43.4°, with its strongest cell at LTDN 12.266 h and pitch 24.8°, where the trajectory reaches a maximum Δa of approximately +20.07 km. Two much smaller retrograde-pitch clusters also appear, but their gains remain below 0.14 km and are negligible relative to the main lobe. Overall, only 89 of 900 sampled cells ($\sim 10\%$) exhibit positive transient SMA increase,~~ 40°; no cell at negative pitch gains, and no cell re-enters within the window. In LTAN the positive cells form one connected band (periodic four-connectivity) running from 20.0 h through midnight to 16.0 h, so ~~non-gain behaviour dominates most~~ 41 of the ~~map.~~ 48 LTAN columns contain a gaining cell; the band is pinched almost to zero near dawn (ridge value 0.46 km at 6.0 h) and absent between 16.5 and 19.5 h (dusk). Along the ridge there are two maxima: the global peak of 5.95 km at (14.0 h, +35°) on the afternoon side and a secondary maximum of 5.75 km at (3.0 h, +40°) on the night side. The most negative 14-day change is $\Delta a(14\,\mathrm{d}) = -29.40$ km at (0.0 h, −30°); drag alone lowers the mean semi-major axis by close to 10 km in 14 days at this altitude (−9.855 km for the drag-only reference run at LTAN 14.0 h, pitch +34°, and −9.89 km for the gain geometry at pitch +35°, the difference being the projected area), so a "neutral" cell is one whose 14-day change is close to that value.

~~The restricted extent of the positive region indicates that the effect is~~ At 700 km the picture is qualitatively different: for the baseline atmosphere none of the 1776 cells shows a positive excursion within 14 days, so that $\Delta a_{\max,14\mathrm{d}}$ vanishes over the whole map, the most negative 14-day change is −62.05 km, and the geometry that peaks at 800 km gives $\Delta a(14\,\mathrm{d}) = -13.75$ km. The regime is therefore geometry-selective~~rather than a generic passive orbit-raising tendency.~~, above all in the sign and magnitude of the pitch (only a sail tilted from the Sun line towards $\hat{\mathbf{s}}_\odot \times \hat{\mathbf{h}}$ by 10° to 60° gains, and the mirror-image tilt always loses), and altitude-selective, since the SRP surplus that the eclipse makes available is worth some 16 km in 14 days (the eclipse on/off separation of Figure 3) and is outweighed by drag 100 km lower. The dependence on LTAN, by contrast, is considerably weaker than the original submission suggested: the band spans most of the LTAN circle, and the two ridge maxima differ by only 0.2 km.

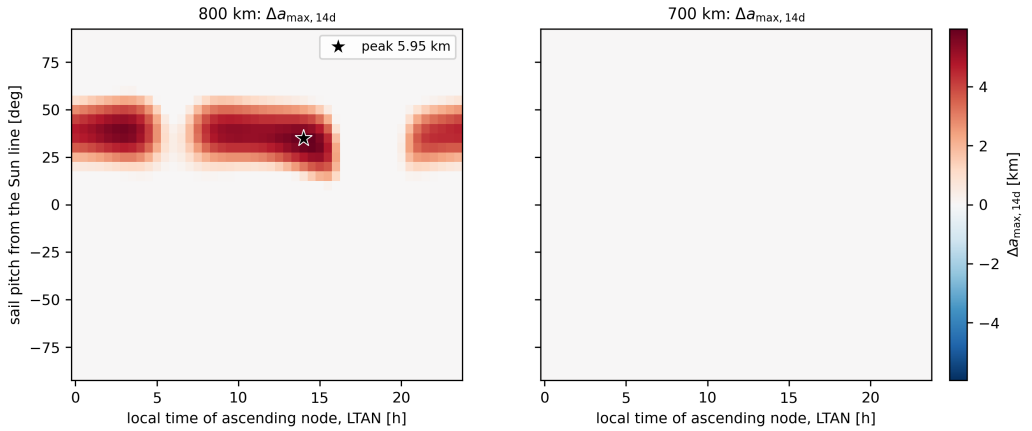


Figure 2: Maximum 14-day excursion of the mean semi-major axis, $\Delta a_{\max,14\mathrm{d}}$, as a function of LTAN and initial sail pitch at 800 km (left) and 700 km (right); $48 \times 37 = 1776$ cells per altitude, RK4 at 10 s, epoch 1 Jan 2026 12:00 UTC, baseline sail of Table 2. At 800 km a single connected band of positive excursion occupies pitch $+10°$ to $+60°$ and LTAN 20.0 h through midnight to 16.0 h, peaking at 5.95 km at LTAN 14.0 h, pitch $+35°$ (star). At 700 km no cell gains.

Three representative ~~700 km~~ 800 km cases are used throughout the remainder of the ~~section: a gain case at RAAN 285.5°, LTDN 12.27 h, and pitch 24.8°; a neutral case at RAAN 111.7°, LTDN 0.68 h, and~~ paper and are collected, together with the former 700 km anchor of the original submission, in Table 5. The gain case is the peak cell of the map, LTAN 14.0 h (RAAN 312.05° at epoch), pitch ~~142.8°; and a decay case at RAAN 99.3°, LTDN 23.85 h, and pitch 155.2°. The gain casereaches a maximum $\Delta a \approx +20.07$ km at approximately day 26.5, but later declines to $\Delta a \approx -13.42$ km by day 75;~~ $+35°$; its 14-day peak falls at day 13.94, the last ascending node of the window, so $\Delta a_{\max,14\mathrm{d}} = \Delta a(14\,\mathrm{d}) = 5.95$ km and the ~~decay case loses SMA monotonically and re-enters by day 42.8~~semi-major axis is still rising at day 14. The decay case is the most negative cell, LTAN 0.0 h (RAAN 102.05°), pitch $-30°$. The neutral case, LTAN 11.0 h (RAAN 267.05°), pitch $+5°$, is a nearly Sun-facing sail in a near-noon plane; it is the no-gain cell with $|\text{pitch}| \leq 45°$ whose 14-day change, -9.885 km, lies closest to the drag-only reference decay of -9.855 km. Over 75 days only the gain case ever rises above a_0; the neutral case ends at -150.9 km, its nearly Sun-facing attitude also presenting a larger projected area to the flow, and the decay case and the former 700 km anchor re-enter. These cases are used to separate bounded

existence from mechanism.

Table 5: Representative cases used in Sections 6–9; the former anchor is the 700 km geometry of the original submission. Epoch 1 Jan 2026 12:00 UTC; inclination 98.31° (800 km) and 98.17° (700 km); $e_0 = 10^{-4}$; $A/m = 10\,\mathrm{m^2\,kg^{-1}}$; $C_r = 1.7$; $C_D = 2.2$; RK4 at 10 s. RAAN is the epoch value corresponding to the stated LTAN (the original submission quoted the former anchor with RAAN 285.5°; 286.04° is the value implied by the LTAN convention of Section 4 at the stated epoch). $\Delta a_{\mathrm{max,14d}}$ is the largest positive excursion of the mean semi-major axis within 14 days (zero if none), $\Delta a(14\,\mathrm{d})$ the value at day 14, $\Delta a_{\mathrm{max,75d}}$ the largest positive excursion within 75 days with its day, and $\Delta a_{75\mathrm{d}}$ the value at day 75, or the re-entry day when the perigee altitude falls below 120 km before then. The 14-day peak of the gain case occurs at day 13.94, the last ascending node of the window.

Case	Alt. [km]	LTAN [h]	Pitch [deg]	RAAN [deg]	$\Delta a_{\mathrm{max,14d}}$ [km]	$\Delta a(14\,\mathrm{d})$ [km]	$\Delta a_{\mathrm{max,75d}}$ [km]	$\Delta a_{75\mathrm{d}}$ [km]
Gain	800	14.0	+35	312.05	5.95	+5.95	+10.86 day 38.1	−2.69
Neutral	800	11.0	+5	267.05	0	−9.89	0	−150.9
Decay	800	0.0	−30	102.05	0	−29.40	0	re-entry day 64.4
Former anchor	700	12.266	+24.8	286.04	0	−21.07	0	re-entry day 39.5

The most direct causal test is to compare the ~~representative~~ gain case with and without eclipse while holding ~~all other initial conditions and model assumptions~~ every other initial condition and model assumption fixed. Figure 3 shows that the short-horizon outcome reverses under that single change. By day 14 ~~,~~ the eclipse-on case reaches ~~$\Delta a \approx +10.31$ km~~ $\Delta a(14\,\mathrm{d}) = +5.95$ km, whereas the eclipse-off counterpart has fallen to ~~$\Delta a \approx -47.75$ km~~ -10.59 km, a separation of ~~approximately 58.06 km. Over the full~~ 16.5 km. The eclipse-off value lies within 0.7 km of the drag-only run of the same geometry (−9.89 km): with the umbra removed, SRP contributes essentially nothing over 14 days, because the orbit-averaged SRP work of a fixed-attitude sail under continuous illumination cancels, exactly as the symmetry argument of Section 3 (Eq. (14)) requires. At the former 700 km anchor (right panel) the values are −21.07 km with eclipse on and −45.36 km with it off: the eclipse still removes the unfavourable arc, worth 24.3 km over 14 days, but at this altitude drag wins regardless. Over 75 ~~-day propagation, the~~ days the

eclipse-on ~~case survives to the end of the run with final $\Delta a \approx -13.42\,\mathrm{km}$, whereas the eclipse-off case re-enters~~gain case ends at $-2.69\,\mathrm{km}$ and the eclipse-off counterpart at $-110.5\,\mathrm{km}$ (Section 7). The gain case is therefore not ~~simply~~ SRP-favourable in a generic sense; its transient improvement depends on eclipse.

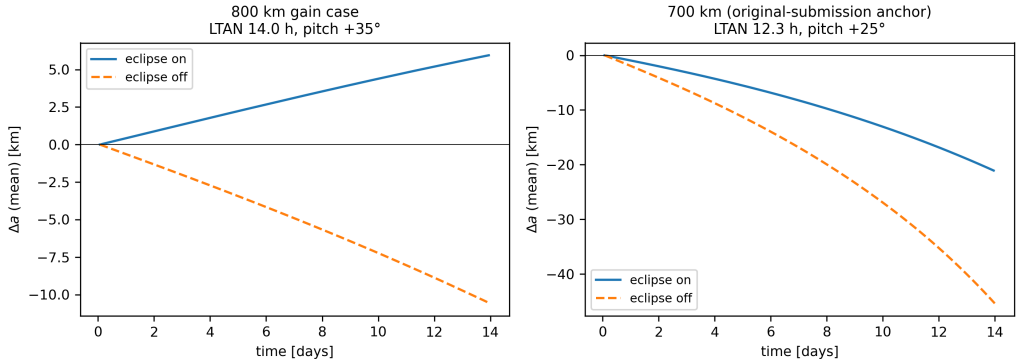


Figure 3: Mean semi-major-axis change over 14 days with eclipse on and off, all other conditions fixed. Left: the 800 km gain case (LTAN 14.0 h, pitch +35°), +5.95 km with eclipse on against −10.59 km with it off. Right: the former 700 km anchor (LTAN 12.266 h, pitch +24.8°), −21.07 against −45.36 km.

A second control isolates which component of the SRP force carries the effect. Repeating the gain case with eclipse on but with the whole SRP force directed along the Sun line instead of the plate normal gives $\Delta a_{\max,14\mathrm{d}} = 0$ and $\Delta a(14\,\mathrm{d}) = -10.50\,\mathrm{km}$: no gain at all, and marginally worse than with SRP switched off ($-9.89\,\mathrm{km}$), because a force of fixed inertial direction along the Sun line performs no net orbit-averaged work in this geometry yet still pumps the eccentricity, exposing the perigee to slightly more drag. This confirms numerically the statement of Sections 3 and 4 that the along-normal (specular) component carries the eclipse-gated mechanism.

The eclipse-on/off comparison establishes causality ~~,~~ but not the orbit-scale mechanism. Figure 4 addresses that ~~mechanism~~ by plotting the instantaneous SRP specific power, ~~$\mathbf{a}_{\mathrm{SRP}}\cdot\mathbf{v}$, over one periapsis-to-periapsis orbit for the representative~~ $\mathbf{a}_{\mathrm{SRP}}\cdot\mathbf{v}$, against time from the ascending node over one node-to-node orbit at epoch (nodal period 100.8 min in all three cases, sampled at 2 s) for the gain, neutral ~~,~~ and decay cases, ~~together with the eclipse interval and a dashed~~ with the umbra shaded and the no-eclipse ref-

erence dashed. In the gain case ~~, eclipse~~ the umbra occupies 32.1% of the orbit and falls predominantly on ~~a would-be negative-power arc : 85.7% of~~ the arc where the normal-directed force would decelerate the sail: 77.4% of the eclipsed samples correspond to negative no-eclipse SRP power, and by integrated magnitude ~~eclipse suppresses roughly $-860\,\mathrm{J\,kg^{-1}}$ of harmful SRP~~ the shadow suppresses $-344.2\,\mathrm{J\,kg^{-1}}$ of harmful work while removing only ~~about $+15\,\mathrm{J\,kg^{-1}}$~~ $+35.3\,\mathrm{J\,kg^{-1}}$ of beneficial work. In the ~~neutral and decay cases, the sign structure reverses: 95.2% and 85.7% of eclipsed samples respectively correspond to would-be *positive* SRP power , so eclipse suppresses the recovery arc instead~~decay case the umbra fraction is 34.9% but only 30.7% of the eclipsed samples are negative: the shadow now covers the accelerating arc, suppressing $+410.2\,\mathrm{J\,kg^{-1}}$ of beneficial work against $-94.4\,\mathrm{J\,kg^{-1}}$ of harmful work. In the neutral case (umbra fraction 34.6%) the SRP power is close to antisymmetric about the umbra, so the shadow removes nearly equal shares of each sign (53.3% of eclipsed samples negative; suppressed work -337.4 against $+264.9\,\mathrm{J\,kg^{-1}}$) and eclipse changes little. The mechanism is therefore sign-selective: the outcome depends on which branch of the SRP-power cycle ~~is removed by shadow.~~

~~Figure 4 therefore identifies the mechanism at the instantaneous level: eclipse changes the sign of the orbit-integrated SRP contribution by removing different branches of the SRP-power cycle in different geometries~~the shadow removes, which is set by the orientation of the umbra relative to the fixed sail normal, i.e. by LTAN and pitch together.

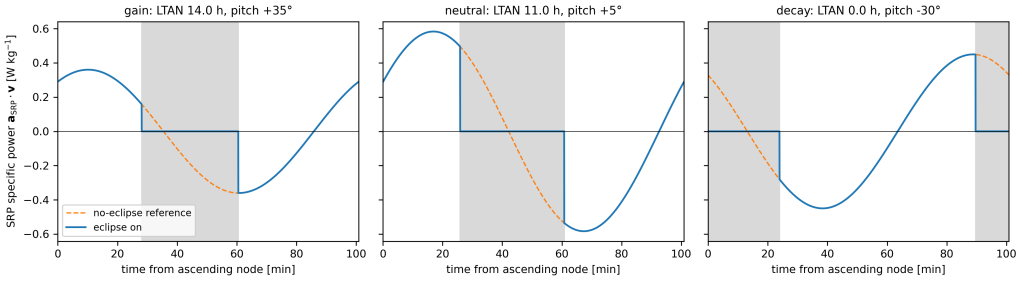


Figure 4: Instantaneous SRP specific power, $\mathbf{a}_{\mathrm{SRP}} \cdot \mathbf{v}$, against time from the ascending node over one node-to-node orbit at epoch (nodal period 100.8 min) for the 800 km gain, neutral and decay cases of Table 5. Shading marks the umbra; the dashed curve is the no-eclipse reference. The shadow suppresses a predominantly negative-power arc in the gain case (77.4% of eclipsed samples negative), a predominantly positive-power arc in the decay case (30.7%), and nearly equal shares in the neutral case (53.3%).

Figure 5 shows the same effect in cumulative form. ~~In the gain case, the eclipse-on orbit closes with~~ With eclipse off, the cumulative SRP specific work ~~of $+799.29\,\mathrm{J\,kg^{-1}}$, compared with $-9.69\,\mathrm{J\,kg^{-1}}$ for the no-eclipse reference. In the~~ over the orbit is near zero in every geometry, $+1.4$, $+0.9$ and $+0.4\,\mathrm{J\,kg^{-1}}$ for the gain, neutral and decay cases~~, the eclipse-on end values are instead $-665.25\,\mathrm{J\,kg^{-1}}$~~: this is the passive cancellation of Section 3, and ~~$-857.93\,\mathrm{J\,kg^{-1}}$, while the corresponding no-eclipse references end near zero at $-4.81\,\mathrm{J\,kg^{-1}}$ and $-53.90\,\mathrm{J\,kg^{-1}}$. The cumulative view therefore confirms the sign-selective interpretation inferred from Figure 4: eclipse enhances the one-orbit SRP balance only~~ it holds irrespective of LTAN and pitch. With eclipse on, the orbit closes at $+310.3\,\mathrm{J\,kg^{-1}}$ in the gain geometry~~and degrades it~~, $+73.5\,\mathrm{J\,kg^{-1}}$ in the neutral ~~and decay geometries.~~

~~The representative gain case also confirms an important limitation of the mechanism. Although the trajectory attains a substantial positive semi-major-axis excursion within the analysed window, that behaviour is not equivalent to clean or sustained whole-orbit raising over longer horizons. The detailed longer-horizon evolution~~ geometry (a quarter of the gain value) and $-315.4\,\mathrm{J\,kg^{-1}}$ in the decay geometry. Eclipse timing therefore sets the *sign* of the orbit-averaged SRP work, not merely its magnitude. The Sun-line variant of the gain case ~~is examined in Section 7~~(dotted curve in the gain panel) closes at $+1.0\,\mathrm{J\,kg^{-1}}$ with eclipse on, indistinguishable from the no-eclipse reference: this is the one-orbit counterpart of the 14-day Sun-line result above. The shadow thus converts a cancelling SRP work cycle into a net surplus only in the gain geometry, into a net deficit in the decay geometry, and in the neutral geometry leaves a small residual that drag swamps.

Taken together, the results in this section establish three points. First, a bounded transient-gain region exists at ~~700 km in LTDN–pitch space~~800 km in LTAN–pitch space, confined to positive pitch between $+10°$ and $+60°$ and peaking at 5.95 km in 14 days, while the same sail gains nowhere at 700 km. Second, eclipse is causally decisive: removing ~~it reverses the short-horizon outcome of the representative gain case~~the umbra at the gain geometry turns +5.95 km into −10.59 km, within 0.7 km of the drag-only outcome of −9.89 km. Third, the ~~orbit-resolved SRP budget identifies the mechanism : eclipse~~ mechanism is sign-selective: under continuous illumination the one-orbit SRP work cancels in every geometry, and the shadow helps only

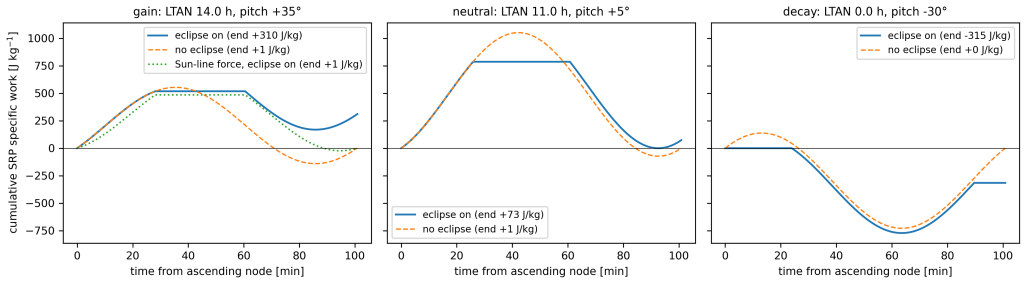


Figure 5: Cumulative SRP specific work over the same node-to-node orbit as Figure 4 for the gain, neutral and decay cases, with eclipse on (solid) and the no-eclipse reference (dashed). The eclipse-on end values are +310.3, +73.5 and −315.4 J kg^{-1}; the no-eclipse references end at +1.4, +0.9 and +0.4 J kg^{-1}. The dotted curve in the gain panel is the Sun-line force variant with eclipse on, ending at +1.0 J kg^{-1}.

when it removes ~~a~~ the negative-power branch of the cycle; the along-normal force component carries the effect, and a Sun-line force of the same magnitude produces none. The gain is nonetheless transient. The 5.95 km attained within 14 days grows to +10.86 km by day 38.1, but this is not clean or sustained whole-orbit raising: by day 75 the mean semi-major axis has fallen back to −2.69 km, through apogee pumping with perigee erosion and the drift of the inertially fixed sail normal away from the Sun line, as Section 7 shows. Nor is the regime an artefact of this implementation: at the gain anchor STK Astrogator gives +6.44 km with eclipse on and −10.13 km with it off over 14 days, against +5.85 and ~~harms when it removes a positive-recovery branch. The representative gain case is therefore physically real, but not a claim of sustained orbit raising.~~ −10.93 km for the present propagator under the centred one-period-mean definition used in that comparison, and Orekit gives +6.13 km against +5.93 km for the same case at a common 30 s step (Section 9).

7. Coupled Orbital Evolution and Breakdown of the Gain Regime

The representative gain trajectory must now be examined over ~~longer horizons~~ a longer horizon to determine what the early increase in semi-major axis actually corresponds to dynamically, and why that favourable behaviour does not persist. The central result is ~~linear~~simple to state: the early increase in a is real, but it is produced mainly by apogee pumping rather than by a clean upward translation of the whole orbit~~; this~~. This increases

eccentricity, lowers perigee ~~,~~ and concentrates drag loss near perigee~~, and eventually causes the late-window~~ ; at the same time the inertially fixed sail normal drifts steadily out of alignment with the Sun, so that the SRP forcing which created the gain collapses in magnitude. The second of these effects dominates the late-time behaviour. Together they drive the orbit-averaged energy balance ~~to turn negative~~ negative at the peak of a, and by day 75 the accumulated surplus has been fully erased.

Figure 6 makes the first part of that breakdown visible directly. For the representative ~~700 km~~ 800 km gain case introduced in Section 6 ~~,~~ (LTAN 14.0 h, pitch $+35°$, RAAN $312.05°$), the 75-day histories of mean a, apogee altitude, perigee altitude ~~,~~ and eccentricity show that the orbit is not lifted uniformly. ~~Instead, apogee grows much more strongly~~ Apogee grows far more than perigee is preserved, so the early gain develops together with a progressive departure from circularity. The neutral and decay cases are overlaid for comparison: the decay case (LTAN 0.0 h, pitch $-30°$) loses altitude monotonically and re-enters on day 64.4, while the neutral case follows an intermediate, drag-like path between the two. Table 6 collects the corresponding quantities for the gain case.

~~The~~ The mean semi-major axis ~~begins at $a_0 = 7{,}078.137$ km, rises early, reaches a maximum of 7,098.211 km, and~~ of the gain case begins at $a_0 = 7178.168$ km, rises to $\Delta a(14\,\mathrm{d}) = +5.95$ km at the last ascending node of the 14-day window (day 13.94; this is the same trajectory, and the same value, that anchors Section 6), and continues upward more slowly: $+9.96$ km on day 28 and a peak of $\Delta a_{\max,75\mathrm{d}} = +10.86$ km ($a = 7189.030$ km) on day 38.1. It then declines for the remainder of the propagation~~to a final value below its initial level~~, passing $+10.70$ km on day 42, $+7.59$ km on day 56 and $+0.59$ km on day 70, returning to its initial value on day 71.0 and finishing at $\Delta a_{75\mathrm{d}} = -2.69$ km ($a = 7175.479$ km). The behaviour is therefore explicitly non-monotonic: there is a finite interval of positive excursion lasting roughly ten weeks, but not sustained ~~whole-horizon~~ orbit raising. ~~In other words, the~~ The short-horizon gain is real, yet ~~it does not remain self-maintaining over the full~~ by the end of the 75-day window it has been entirely given back.

The apogee–perigee split shows how that ~~breakdown~~ reversal develops. Apogee ~~rises strongly and remains elevated, whereas perigee falls substantially and remains depressed. At the same time, the eccentricity~~

Table 6: Long-horizon summary of the representative 800 km gain case over the 75-day propagation (1069 orbits). The case corresponds to altitude 800 km, LTAN 14.0 h (RAAN 312.05°), pitch +35°, inclination 98.31°, $e_0 = 10^{-4}$, sail area 100 m^2, mass $m = 10$ kg, $C_r = 1.7$ and $C_D = 2.2$. Mean (secular) elements are osculating values sampled at ascending-node crossings. All work quantities are specific, i.e. per unit mass ($m = 10$ kg).

Quantity	Initial	Peak / minimum (day)	Final (day 75)
Mean semi-major axis, a [km]	7178.168	7189.030 (max, day 38.1)	7175.479
Peak excursion, $\Delta a_{\max,75\mathrm{d}}$ [km]	+10.86 (day 38.1)		
Net change, $\Delta a_{75\mathrm{d}}$ [km]	−2.69		
Apogee altitude [km]	801.0	912.0 (max, day 43.9)	884.7
Perigee altitude [km]	799.1	708.6 (min, day 48.4)	709.9
Eccentricity, e	1.35×10^{-4}	1.41×10^{-2} (max, day 45.2)	1.22×10^{-2}
Cumulative SRP work at day 75 [kJ kg^{-1}]	+194.5		
Cumulative drag work at day 75 [kJ kg^{-1}]	−204.4		
Cumulative total work at day 75 [kJ kg^{-1}]	−9.9		
Work ratio at day 75, $\lvert W_{\mathrm{drag}}\rvert \,/\, \lvert W_{\mathrm{SRP}}\rvert$	1.051		

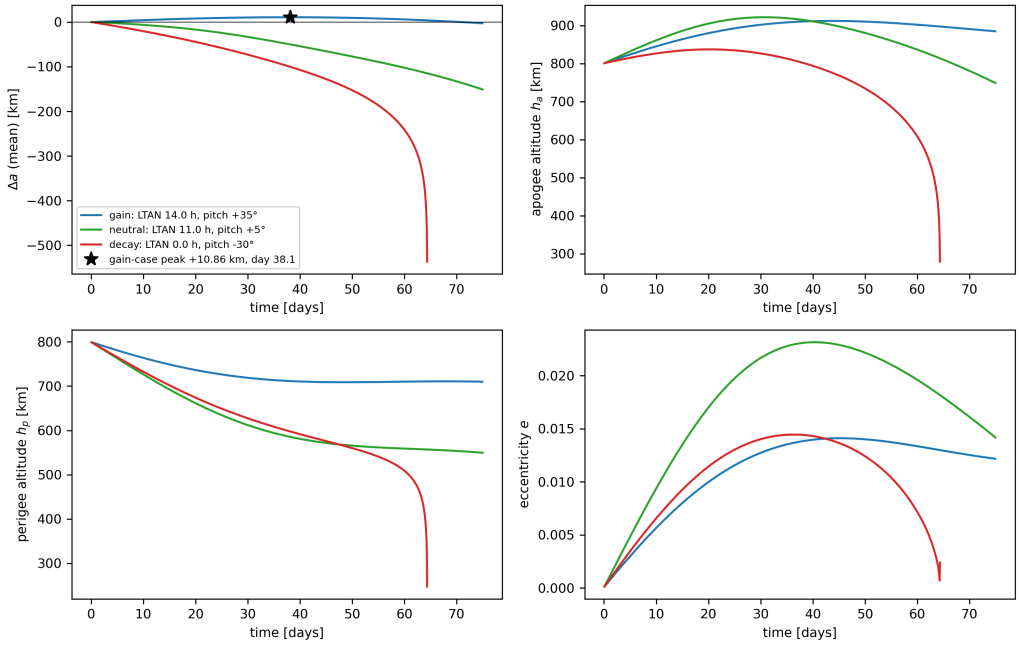


Figure 6: Long-horizon evolution of mean orbital elements over 75 days for the representative 800 km gain, neutral and decay cases: change in mean semi-major axis Δa (with the gain-case peak marked), apogee altitude, perigee altitude and eccentricity. The gain case shows non-monotonic semi-major-axis evolution, strong apogee growth, perigee erosion and eccentricity growth; the decay case re-enters on day 64.4.

~~grows~~ altitude rises from ~~an initially near-circular value to an order 10^{-2} state~~ 801.0 km to 860.4 km by day 14 and 910.4 km by day 38, peaks at 912.0 km on day 43.9, and is still 884.7 km at day 75. Perigee altitude falls from 799.1 km to 751.6 km by day 14 and 711.3 km by day 38, reaches its minimum of 708.6 km on day 48.4, and remains depressed at 709.9 km at day 75. The eccentricity correspondingly grows from 1.35×10^{-4} to a maximum of 1.41×10^{-2} on day 45.2 and relaxes only ~~partially thereafter~~slightly to 1.22×10^{-2}. The key point is ~~therefore~~ geometric rather than merely descriptive: the ~~early increase~~ +10.86 km peak in a is achieved ~~primarily~~ by stretching the orbit, with apogee raised by over 100 km while perigee is lowered by about 90 km, not by lifting it uniformly.

The physical consequence is that drag loss becomes progressively concentrated near perigee, where the atmosphere is densest. This is the ~~clearest~~ first breakdown pathway visible in the long-horizon outputs: the same evolution that produces the early gain also creates a lower-perigee, more eccentric orbit that is increasingly ~~vulnerable~~ exposed to atmospheric loss. ~~No stronger claim than that is needed here.~~ Eclipse remains essential throughout. Repeating the gain case with the umbra disabled, so that SRP acts around the whole orbit, gives $\Delta a_{75\mathrm{d}} = -110.5$ km, a perigee minimum of 592.7 km and $e_{\max} = 0.0189$: without the eclipse-induced sign selectivity of Section 6 the same attitude pumps eccentricity even harder while contributing essentially no net energy, and the orbit ends 108 km lower after 75 days. The decay case illustrates the same eccentricity pumping with the wrong sign of net work: its eccentricity reaches 0.0145 on day 36.6, and perigee erosion then runs away to re-entry on day 64.4. The former 700 km configuration of the original submission (LTAN 12.266 h, pitch 24.8°) behaves likewise and re-enters on day 39.5.

The cumulative work budget in Figure 7 must be read carefully. Over the full 75-day window ~~, cumulative SRP work remains slightly larger than~~ the cumulative specific SRP work is $+194.5\,\mathrm{kJ\,kg^{-1}}$ and the cumulative drag work is $-204.4\,\mathrm{kJ\,kg^{-1}}$, so the cumulative total is ~~still marginally positive by day 75. That does *not* mean the regime is still~~ $-9.9\,\mathrm{kJ\,kg^{-1}}$ and $|W_{\mathrm{drag}}|/|W_{\mathrm{SRP}}| = 1.051$. The total does not fall monotonically: it rises to a maximum of $+41.3\,\mathrm{kJ\,kg^{-1}}$ on day 38.1, the same day on which mean a peaks, and then declines, crossing zero on day 71.2. This is consistent with the semi-major-axis history, since $\Delta a \simeq (2a^2/\mu)\,W_{\mathrm{total}}$ gives -2.6 km for the day-75 total, matching the node-sampled $\Delta a_{75\mathrm{d}} = -2.69$ km and the

zero crossing of mean a on day 71.0. The regime therefore does more than stop strengthening at late times~~. It means only that an earlier accumulated surplus has not yet been fully erased. By the end of the run, that surplus is nearly exhausted, with $|W_{\mathrm{drag}}|/|W_{\mathrm{SRP}}| \approx 0.9961$~~: the surplus accumulated during the favourable phase is unwound in full, and slightly overshot, within the analysed window.

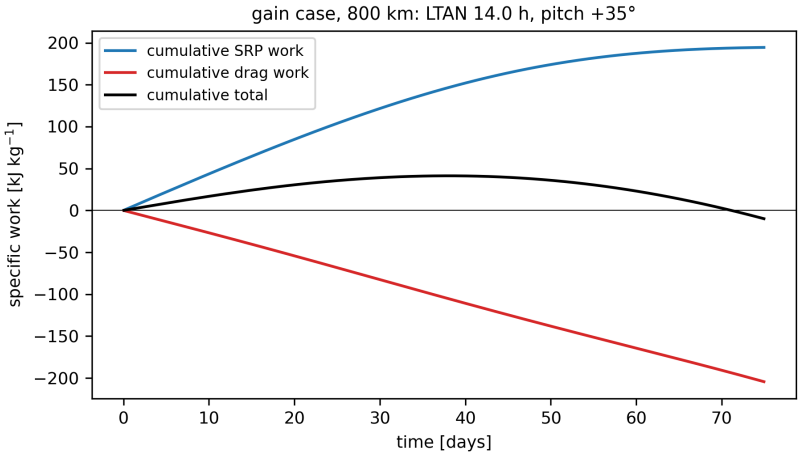


Figure 7: Cumulative specific SRP work, drag work and total work ($\mathrm{kJ\,kg^{-1}}$) for the representative 800 km gain case over 75 days. The cumulative total peaks at $+41.3\,\mathrm{kJ\,kg^{-1}}$ on day 38.1 and turns negative on day 71.2; by day 75 cumulative drag exceeds cumulative SRP work by 5.1%.

The more revealing diagnostic is ~~therefore~~ the orbit-averaged early-versus-late comparison in the left panel of Figure 8 and in Table 7, which compares the first and last 20% of the run (214 of 1069 orbits each). Early in the run ~~,~~ the mean SRP contribution of $+301.3\,\mathrm{J\,kg^{-1}\,orbit^{-1}}$ exceeds the mean drag loss of $-188.3\,\mathrm{J\,kg^{-1}\,orbit^{-1}}$, so the per-orbit balance is ~~positive~~$+113.0\,\mathrm{J\,kg^{-1}\,orbit^{-1}}$. Late in the run ~~,~~ the mean SRP contribution has collapsed ~~while drag remains large, and the orbit-averaged net balance becomes negative. The breakdown is therefore already present on a local~~ to $+33.2\,\mathrm{J\,kg^{-1}\,orbit^{-1}}$, a factor of 0.110 of its early value, while the drag loss is essentially unchanged at $-187.6\,\mathrm{J\,kg^{-1}\,orbit^{-1}}$ (ratio 0.996); the net balance is $-154.4\,\mathrm{J\,kg^{-1}\,orbit^{-1}}$. The per-orbit ~~basis even though the full-window cumulative total remains slightly positive.~~

~~The correct long-horizon interpretation is therefore a timescale split: the trajectory retains a small cumulative surplus from its earlier favourable~~

~~phase, but its late-time orbit-averaged balance has already turned negative. The regime is not sustained; it is being unwound.~~

~~The physical picture is therefore one of temporary outperformance, not eliminated decay~~ net work first turns negative on day 38.1, at orbit 543, which is exactly the peak of mean a: the peak is simply the point at which the local balance changes sign, and everything after it is decay. Two features of this comparison deserve emphasis. First, the drag loss per orbit is almost identical in the two windows, so the deterioration of the balance is driven entirely by the loss of SRP work rather than by growth of the drag penalty; perigee erosion matters because it concentrates that penalty near perigee and leaves the orbit less able to tolerate the loss, as the decay case shows, not because it enlarges the penalty within this window. Second, the eclipse fraction per orbit barely changes, 0.323 early against 0.330 late, so the loss of SRP work is not a consequence of changing eclipse geometry. Its cause lies in the attitude. ~~Eclipse-modulated SRP produces an early advantage, but the resulting orbital distortion lowers perigee, increases eccentricity, and makes subsequent drag losses harder to offset. The mechanism delays deterioration for a finite interval; it does not remove the long-horizon tendency toward decay.~~

The ~~longer-term weakening is also consistent with the~~ late-time weakening follows directly from the SRP model and the force-decomposition framework ~~introduced in Chapter~~of Section 4. There, using the Gauss Variational Equations, we argued that semi-major-axis growth depends ~~primarily~~ on maintaining a favourable in-plane ~~force, especially~~ forcing, in particular a strong transverse component F_T, while ~~avoiding excessive projection into the normal direction F_N, which does not contribute directly to useful in-plane energygrowth. The representative gain case shows a major redistribution of the SRP force between these channels. As shown in Figure 9,~~ the ~~orbit-mean magnitude of the transverse component decreases from approximately 3.529×10^{-4} N early in the mission to approximately 7.312×10^{-5} N late in the mission, a reduction~~ normal component F_N contributes nothing directly to the energy. The sail normal is held fixed in inertial space, but the Sun moves along the ecliptic at roughly 1° per day and the Sun-synchronous orbit plane precesses with it, so the Sun–normal angle α opens steadily: from 35.0° at day 0 to 39.4° on day 14, 48.6° on day 30, 54.2° on day 38, 59.4° on day 45, 71.2° on day 60 and 83.4° on day 75. Since the specular force scales with $\cos^2 \alpha$, which falls from 0.671 to 0.013 over the run, the

Table 7: Early-versus-late comparison for the representative 800 km gain case. The orbit-averaged work budget and eclipse fraction use 20% windows (first and last 214 of 1069 orbits); the orbit-mean SRP acceleration components use 10% windows (first and last 107 orbits), matching the conventions of the figure-generation scripts. $|a_T|$ and $|a_N|$ are the transverse and normal components of the SRP acceleration in the local orbital frame of Section 4; α is the angle between the Sun line and the inertially fixed sail normal, quoted at day 0 and day 75.

Metric	Early window	Late window	Late / Early
*Orbit-averaged work budget (*20% *windows;* 214 *orbits each)*			
Mean SRP work [J kg^{-1} orbit^{-1}]	+301.3	+33.2	0.110
Mean drag work [J kg^{-1} orbit^{-1}]	−188.3	−187.6	0.996
Mean net work [J kg^{-1} orbit^{-1}]	+113.0	−154.4	—
Eclipse fraction per orbit	0.323	0.330	—
*SRP acceleration projection (*10% *windows;* 107 *orbits each)*			
Orbit-mean $\lvert a_T \rvert$ [μm s^{-2}]	21.6	1.08	0.050
Orbit-mean $\lvert a_N \rvert$ [μm s^{-2}]	17.0	1.49	0.088
Ratio $\lvert a_N \rvert / \lvert a_T \rvert$	0.79	1.38	—
Sun–normal geometry (day 0 *and day* 75*)*			
Sun–normal angle, α [deg]	35.0	83.4	—
$\cos^2 \alpha$	0.671	0.013	—

SRP acceleration on the sail is reduced by a factor of about ~~4.8. Over the same early-to-late comparison, the~~ fifty. The right panel of Figure 8 shows the result in the orbital frame: over the first and last 10% windows (107 orbits each) the orbit-mean ~~magnitude of the normal component increases from approximately 3.224×10^{-5} N to approximately 1.727×10^{-4} N, an increase by~~ transverse component $|a_T|$ falls from 21.6 to 1.08 μm s^{-2}, a factor of 0.050, and the orbit-mean normal component $|a_N|$ falls from 17.0 to 1.49 μm s^{-2}, a factor of ~~about 5.4.~~ 0.088. The SRP force therefore does not merely redistribute between the transverse and normal channels; its magnitude collapses, and what remains projects relatively more into the normal channel, with $|a_N|/|a_T|$ rising from 0.79 to 1.38. The transverse forcing that generates $\dot{a}$ is thus doubly penalised, by the overall loss of illumination and by the less favourable projection of what is left.

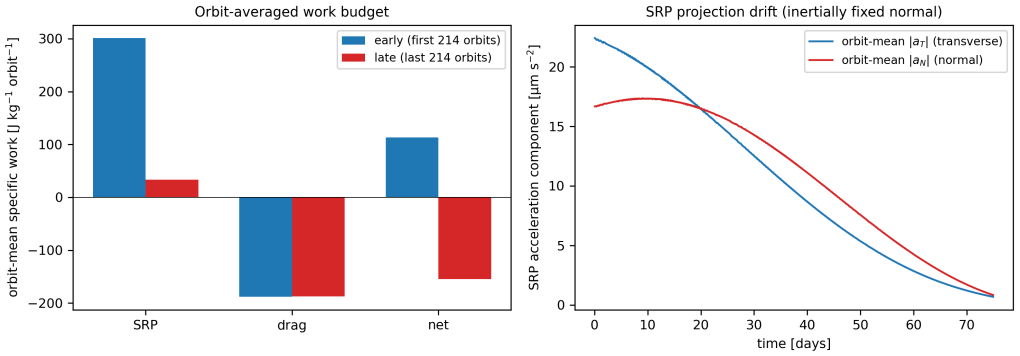


Figure 8: Early-versus-late diagnostics for the representative 800 km gain case. Left: orbit-averaged specific SRP, drag and net work (J kg^{-1} orbit^{-1}) in the first and last 20% windows (214 orbits each); the SRP contribution collapses while the drag penalty is unchanged, so the late-window net balance is negative. Right: orbit-mean transverse $|a_T|$ and normal $|a_N|$ components of the SRP acceleration (μm s^{-2}) as functions of time; both fall by more than an order of magnitude as the inertially fixed sail normal drifts away from the Sun line.

This ~~redistribution of SRP force~~ attitude drift is not secondary bookkeeping~~. Because the sail orientation is~~ ; it is the dominant late-time effect. Figure 9 sketches the geometry. A spin-stabilised sail with its normal fixed in inertial space is well aligned with the Sun at the start of the run, when the pitch of +35° places the SRP force favourably in the local RT plane during the sunlit arc. Two months later the Sun has moved some 60° along the ecliptic

and the orbit plane has precessed with it, while the ~~orbit evolves, sail~~ normal has not moved at all. The same inertially fixed attitude therefore sees the Sun at a grazing angle and projects what little force remains less favourably into the local frame. The mechanism established in Section 6 requires both the eclipse-induced sign selectivity and a sail that faces the Sun; the first is preserved over 75 days, but the second is lost by any attitude that is not actively steered. This is the ~~SRP force becomes progressively less well aligned with the favourable local RT-plane direction. The transverse component weakens strongly, the normal component becomes relatively larger on an early-to-late basis, and the contribution to $\dot{a}$ correspondingly degrades. This geometrical drift is therefore a~~ second breakdown pathway, ~~alongside the drag concentration caused by perigee erosion~~and in this case the stronger one.

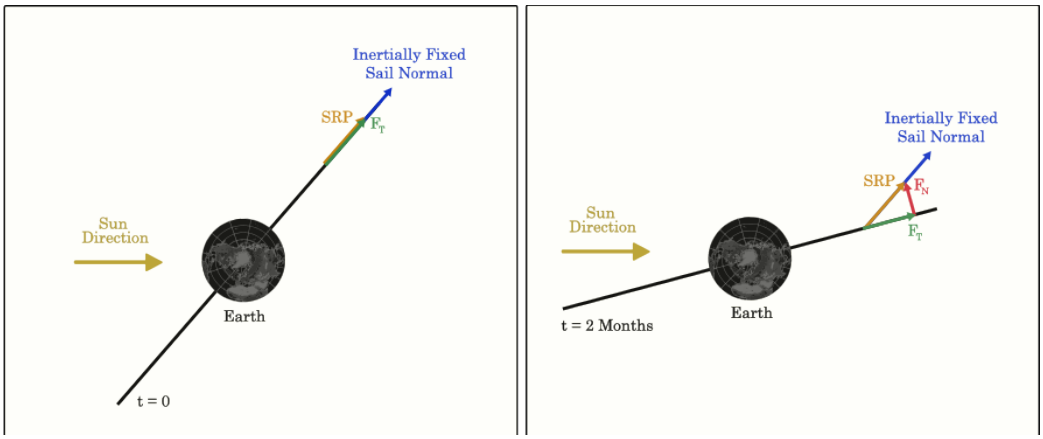


Figure 9: Illustration of SRP projection drift for an inertially fixed, spin-stabilised sail in a precessing Sun-synchronous orbit. At $t = 0$ (left), the fixed sail normal is 35° from the Sun line and produces a favourable transverse SRP component F_T. After two months (right), the Sun has moved along the ecliptic and the orbit plane has precessed with it, so the same inertially fixed attitude is illuminated at a grazing angle and projects less favourably into the local frame, with a much reduced F_T and a relatively larger normal contribution F_N. This geometric drift is the dominant cause of the breakdown of the transient gain regime over the 75-day horizon.

Taken together, the results in this section resolve the apparent tension between early gain and later decay. The mechanism established in Section 6 is real, but it is front-loaded and self-limiting. ~~Early~~ The early positive Δa is achieved mainly through apogee pumping rather than clean whole-orbit lifting; that evolution increases eccentricity and lowers perigee by some 90 km,

which concentrates drag loss, while the ~~fixed inertial sail orientation also~~ inertially fixed sail orientation causes the SRP ~~force to drift away from the most favourable transverse direction. The late-time~~ acceleration to shrink by a factor of about fifty as the Sun–normal angle opens from 35.0° to 83.4°. The orbit-averaged balance ~~therefore turns negative even though a small cumulative surplus from the earlier phase remains~~ turns negative on day 38.1, at the peak $\Delta a_{\mathrm{max,75d}} = +10.86\,\mathrm{km}$; mean a returns to its initial value on day 71.0; and by day 75 the cumulative total work is $-9.9\,\mathrm{kJ\,kg^{-1}}$ with $\Delta a_{75\mathrm{d}} = -2.69\,\mathrm{km}$. The regime does not fail abruptly; it progressively exhausts the margin that created it~~.~~, and within the analysed window it exhausts that margin completely. How sensitive this behaviour is to the geometric and physical parameters of the case is the question taken up in Section 8.

8. Robustness and sensitivity of the gain-anchor case

To assess whether the ~~identified~~ short-horizon gain ~~behaviour~~ identified in Section 6 is a numerically isolated point or instead occupies a finite~~bounded neighborhood~~, bounded neighbourhood, a set of local sensitivity sweeps was carried out about the same representative gain-anchor geometry over a fixed 14-day analysis window. In all cases ~~,~~ the response metric is the ~~maximum semi-major-axis increase attained~~ largest positive excursion of the mean semi-major axis above its initial value within that window,

$$\Delta a_{\mathrm{max,14d}} = \max\left(0,\ \max_{t\in[0,14\,\mathrm{d}]} \left(a(t) - a_0\right)\right), \tag{39}$$

expressed in kilometres~~.~~, where $a(t)$ is the osculating semi-major axis sampled at ascending-node crossings. The metric is set to zero whenever the mean semi-major axis never rises above a_0; in those cases the value at day 14, $\Delta a(14\,\mathrm{d})$, is quoted instead so that the depth of the decay remains visible. This is therefore not the final semi-major-axis change at day 14, but the largest ~~positive excursion relative to the initial orbit~~ gain attained at any time within the first 14 days.

The common reference geometry for all three sweeps is ~~defined by $\Omega = 285.5°$, LTDN $= 12.266176\,\mathrm{hr}$, nominal pitch $24.8°$~~the 800 km gain anchor of Section 6: LTAN $= 14.0\,\mathrm{h}$ ($\Omega = 312.05°$ at epoch), pitch $+35°$, zero roll~~offset~~, a $10 \times 10\,\mathrm{m}$ sail ~~, and~~ ($A = 100\,\mathrm{m^2}$), $m = 10\,\mathrm{kg}$ and hence

$A/m = 10\,\mathrm{m^2\,kg^{-1}}$, with the same configuration propagated at both . The same configuration is also propagated at 700 km and 800 km. The corresponding (inclination 98.17°) for contrast. The nominal responses are $\Delta a_{\mathrm{max,14d}} = 14.958\,\mathrm{km}$ at 700 km and $33.937\,\mathrm{km}$ at 800 km. The 800 km curves should therefore be interpreted as $\Delta a_{\mathrm{max,14d}} = 5.95\,\mathrm{km}$ at 800 km and zero at 700 km, where the same attitude and node phasing give $\Delta a(14\,\mathrm{d}) = -13.75\,\mathrm{km}$. The 700 km curves in Fig. 10 are therefore a same-geometry altitude comparison rather than as a separately re-optimised gain anchoranchor; with the nominal atmosphere, no roll offset and no area-to-mass ratio in the sampled ranges produces a gain at 700 km.

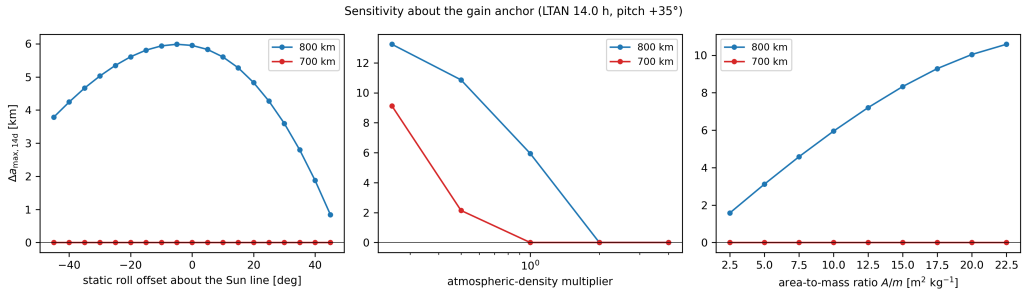


Figure 10: Sensitivity of the 14-day maximum semi-major-axis increase, $\Delta a_{\mathrm{max,14d}}$, about the 800 km gain anchor (LTAN 14.0 h, $\Omega = 312.05°$, pitch +35°, $A/m = 10\,\mathrm{m^2\,kg^{-1}}$), with the same geometry propagated at 700 km in each panel. Left panel: static roll about the Sun line from −45° to +45°. Centre panel: multiplicative scaling of the US-1976 density profile from 0.25× to 4× (logarithmic axis). Right panel: area-to-mass ratio from 2.5 to $22.5\,\mathrm{m^2\,kg^{-1}}$ with the mass fixed at 10 kg. The 700 km response is zero everywhere except at reduced density (0.25× and 0.5×); at 4× density the 700 km sail re-enters within the window and no value is plotted.

8.1. ***Static roll sensitivity***

At the same time, the envelope remains clearly bounded at both altitudes. As the roll magnitude increases beyond roughly 20°, the attainable gain decreases steadily, showing that the eclipse-biased work surplus is still alignment-sensitive. The important additional result is that the sensitivity weakens markedly at higher altitude. The left panel of Fig. 10 shows the response to a static roll offset about the Sun line, sampled at 5° intervals from −45° to +45°. At 700 km, offsets of ±45° retain only a small fraction of the zero-roll peak, whereas at 800 km the same offsets still preserve

~~roughly~~ envelope is broad: the maximum of 5.99 km occurs at −5° rather than at zero roll, but it lies within 0.7% of the nominal 5.95 km, so the anchor is effectively at the crest of the roll response. The fall-off is markedly asymmetric. Towards negative roll the response declines slowly, from 5.94 km at −10° through 5.03 km at −30° to 3.79 km at −45°, where 64% of the nominal response remains. Towards positive roll it collapses: 5.28 km at +15°, 3.60 km at +30°, 1.88 km at +40° and only 0.84 km at +45°, 14% of the nominal value. More than half of the nominal response ~~. In absolute terms, the 800 km case with a 45° roll offset still produces a gain comparable to the zero-roll nominal case at~~ is retained over the whole negative-roll side sampled (64% still remains at −45°) and up to roughly +34° on the positive side, so the admissible alignment band is at least 79° wide and displaced towards negative roll. The asymmetry is a consequence of the one-sided plate and the signed in-plane tilt: a roll of one sign moves the sail normal towards the orbit normal on the side that keeps the illuminated arc favourably oriented, whereas the opposite sign moves it away and reduces the eclipse-biased work surplus more quickly than it reduces the projected drag area.

At 700 km~~. The sweep remains mildly asymmetric about zero, consistent with~~ , $\Delta a_{\mathrm{max,14d}} = 0$ for every roll offset. The decay at day 14 ranges from −8.65 km at −45° to −23.7 km at +45°, with −13.75 km at zero roll. Rolling the sail away from the Sun line therefore reduces the projected drag area more than it reduces the SRP work on the negative-roll side, which slows the decay, but at no sampled roll is the ~~present one-sided SRP illumination model, but the dominant result is not the asymmetry itself; it is that the admissible roll envelope broadens substantially once the drag environment is relaxed.~~ reduction sufficient to turn the net 14-day change positive. The roll sweep thus shows a wide alignment tolerance at 800 km and no admissible roll at all at 700 km under the nominal atmosphere.

8.2. ***Sensitivity to atmospheric density level***

~~Figure 12~~ The centre panel of Fig. 10 shows the effect of scaling the ~~atmospheric density model~~ US-1976 density profile by a constant multiplicative factor while holding the ~~gain-anchor geometry fixedat both 700 km and 800 km. At 700 km,~~ anchor geometry fixed. At 800 km the nominal atmosphere ~~case~~ (1×) gives ~~$\Delta a_{\mathrm{max,14d}} = 14.96$ km, which increases to 28.59 km~~ $\Delta a_{\mathrm{max,14d}} = 5.95$ km, which rises to 10.86 km at 0.5× density

and ~~34.89 km~~ 13.24 km at 0.25×. ~~Increasing density , however, suppresses the gain completely at this altitude~~Doubling the density is already sufficient to suppress the gain entirely: at 2× ~~and~~ the mean semi-major axis never rises above a_0 and $\Delta a(14\,\mathrm{d}) = -4.53\,\mathrm{km}$, while at 4× ~~density, $\Delta a_{\mathrm{max,14d}} = 0$. At 800 km, by contrast, the same sweep remains positive throughout the full tested range, with 33.94 km at nominal density, 26.49 km at~~ the decay reaches −28.8 km. At 700 km the gain exists only in the thinned atmosphere, with 9.13 km at 0.25× and 2.14 km at 0.5×; at 1× and 2× ~~, density~~ the response is zero with $\Delta a(14\,\mathrm{d}) = -13.75\,\mathrm{km}$ and ~~10.25 km even~~ −59.7 km respectively, and at 4× ~~density.~~ the sail re-enters before the end of the 14-day window.

This sweep shows most clearly that drag is the ~~dominant~~ bounding mechanism, ~~but it also shows that the location of the~~ and it locates the drag-limited boundary ~~depends strongly on altitude. At~~ at the anchor geometry: between 1× and 2× the US-1976 density at 800 km, and between 0.5× and 1× at 700 km. In other words, 700 km ~~, the short-horizon semi-major-axis increase exists only while the eclipse-biased SRP work surplus exceeds the accumulated drag losses; once the atmosphere is strengthened sufficiently, the drag penalty dominates from the outset and the semi-major axis never rises above its initial value within the~~ sits just beyond the boundary and 800 km lies within a factor of two of it. The gain at 800 km does not survive the full tested range; it is lost once the density exceeds somewhere between one and two times the nominal profile. This margin matters because the US-1976 profile is a fixed reference rather than a prediction of the actual thermosphere: Section 9 shows that realistic thermospheric models (NRLMSISE-00, Harris–Priester) increase the 14-day ~~window. At 800 km, however, the same mechanism remains much more resilient: increased density degrades the gain substantially, but does not eliminate it across the tested range~~drag decay at 800 km by factors of 1.6 to 3.1 relative to US-1976, so the present sweep brackets that model uncertainty rather than lying comfortably inside it.

~~The most defensible interpretation is therefore not that the regime is uniformly fragile, but that it is strongly altitude-dependent and can lie either near or well away from the drag-limited boundary depending on orbital height.~~

8.3. ***Sensitivity to area-to-mass ratio***

~~Figure 13~~ The right panel of Fig. 10 shows the response to ~~varying~~ the area-to-mass ratio ~~while keeping~~ with the spacecraft mass fixed at 10 kg. The square sail side length is varied from 5 to 15 m, corresponding to A/m ~~values~~ from 2.5 to 22.5 $\mathrm{m^2\,kg^{-1}}$ ~~. At 700 km,~~ in steps of 2.5 $\mathrm{m^2\,kg^{-1}}$. At 800 km the response is ~~non-monotonic. Starting from low A/m, the maximum short-horizon gain rises from 4.17 km at $A/m = 2.5\,\mathrm{m^2\,kg^{-1}}$ to the nominal 14.96 km at $A/m = 10\,\mathrm{m^2\,kg^{-1}}$, then reaches a broad maximum of 17.92 km at $A/m = 14.4\,\mathrm{m^2\,kg^{-1}}$, before declining slightly at larger sail sizes. At 800 km, by contrast, the response remains monotonic over the full sampled range~~monotonic over the whole sampled range, rising from 1.58 km at $A/m = 2.5\,\mathrm{m^2\,kg^{-1}}$ through 3.12, ~~rising from 8.54 km at $A/m = 2.5\,\mathrm{m^2\,kg^{-1}}$ to 71.61 km at $A/m = 22.5\,\mathrm{m^2\,kg^{-1}}$, with no~~ 4.58 and the nominal 5.95 km at 10 $\mathrm{m^2\,kg^{-1}}$, then 7.21, 8.33, 9.29 and 10.04 km, to 10.59 km at 22.5 $\mathrm{m^2\,kg^{-1}}$. The curve is, however, distinctly concave: the increment per 2.5 $\mathrm{m^2\,kg^{-1}}$ step falls from 1.54 km at the low end to 0.54 km at the high end, so diminishing returns are already evident even though no roll-over ~~yet visible.~~ occurs within the sampled range. At 700 km the response is zero for every A/m, and $\Delta a(14\,\mathrm{d})$ deteriorates from −2.61 km at 2.5 $\mathrm{m^2\,kg^{-1}}$ to −60.4 km at 22.5 $\mathrm{m^2\,kg^{-1}}$.

The ~~altitude split again changes the interpretation. At 700 km, the non-monotonicity shows that increasing sail size does not produce indefinite improvement: larger~~ two altitudes read differently for the same reason. Because the sail area enters both the SRP and the drag acceleration, increasing A/m ~~strengthens SRP , but it also strengthens drag and modifies the coupled orbital response over the 14-day window, producing a finite optimum and then a shallow plateau~~scales the two competing effects together: it does not change which of them wins, only how quickly the outcome is reached. At 800 km, ~~however, the same drag-induced roll-over is pushed beyond the sampled range, so increasing A/m continues to improve the short-horizon gainthroughout the tested envelope. The shared conclusion is that stronger~~ where the eclipse-biased SRP work surplus exceeds the drag loss at the anchor geometry, a larger sail raises the gain, but the drag penalty grows alongside it and the concavity of the curve is the first signature of the roll-over that must eventually follow. At 700 km, where drag already wins, a larger sail simply decays faster. Stronger SRP authority is ~~beneficial, but the accessible benefit depends on how strongly drag competes~~ therefore

beneficial only on the gain side of the drag-limited boundary, and even there with diminishing returns over the observation window.

~~Compared with the density sweep, the A/m response remains broad at both altitudes, but only the lower-altitude case shows a clear local optimum within the sampled range.~~

8.4. Interpretation of the sensitivity envelope

Taken together, the three sweeps support a restrained ~~but more nuanced~~ conclusion. The transient gain ~~behaviour~~ at 800 km is not a numerically singular point: it persists over a ~~finite neighborhood in roll and over a broad range of positive~~ roll envelope at least 79° wide (from the −45° sampling limit to roughly +34°) and over the entire sampled range of A/m, and ~~this admissible neighborhood expands significantly between 700 km and 800 km~~the anchor itself lies at the crest of the roll response. At the same time ~~,~~ the mechanism is ~~still~~ not a generally robust passive orbit-raising mode. ~~Its existence remains contingent on favourable Sun-orbit-attitude phasing and on a dragenvironment weak enough for the eclipse-biased SRP work surplus to remain positive over the observation window.~~ It is bounded by drag, and the admissible neighbourhood is strongly altitude dependent: at 700 km the same geometry gains only if the atmospheric density is at most half the US-1976 value, and neither roll nor sail size can recover it under the nominal atmosphere, whereas at 800 km the gain is lost once the density is doubled.

The most defensible statement is therefore that the gain occupies a bounded local envelope whose size ~~depends materially on~~ is set by the margin between the SRP work surplus and the drag loss at the chosen altitude. Roll sensitivity ~~demonstrates finite but limited~~ shows a wide but asymmetric alignment tolerance at ~~700 km, but a substantially wider tolerance band at~~ 800 km. Density sensitivity shows that drag ~~remains~~ is the dominant bounding mechanism ~~, while also showing~~ and that the regime ~~can lie either close to or well away from the drag-dominated breakdown threshold depending on orbital height~~sits within a factor of two of the drag-limited boundary at 800 km and just beyond it at 700 km. Area-to-mass sensitivity shows that stronger SRP authority improves the short-horizon gain ~~broadly, but that the onset of diminishing returnsis itself altitude-dependent and is shifted to larger A/m at higher altitude~~but with diminishing returns, because drag and SRP scale together. These results are consistent with ~~the broader~~

~~thesis of the paper: the mechanism is real and physically interpretable, but it is geometry-bounded, alignment-sensitive, and ultimately limited by the coupled drag and orbital-distortion effects that later drive the~~ Section 7, where the same drag penalty, acting together with the coupled distortion of the orbit, drives the long-horizon transition from transient gain to decay, and they motivate the atmosphere-model dependence examined in Section 9, since the drag-limited boundary at 800 km lies within the span of drag levels (1.6 to 3.1 times the US-1976 decay) that realistic thermospheric models return.

9. Independent numerical validation: an open-source propagator cross-checked against STK Astrogator and Orekit

The ~~reduced-order analysis~~ results of Sections 6–8 ~~establishes the passive-gain regime within a single dynamical model~~were produced by a single open-source propagator written for this class of passive-sail SRP–drag problem. Because the gain regime is narrow and ~~depends~~ rests on a delicate balance between eclipse-gated SRP work and atmospheric drag, its credibility ~~rests~~ depends on two things being demonstrated independently: first, that the gain region *exists* and is not an artefact of one numerical implementation; and second, that the propagator used to map it is itself trustworthy. This section addresses both ~~. We re-implement the model in an independent, open-source propagator and cross-validate~~ by cross-validating that propagator, component by component, against two independent references: Ansys Systems Tool Kit (STK) with the Astrogator module [24, 25], a widely used industry-standard flight-dynamics environment, for the conservative dynamics, the application of drag and, after two configuration corrections described below, the SRP response; and the Orekit library [28] for the force models, the gain regime itself, and the sensitivity of the results to the atmosphere model and the solar ephemeris.

The motivation is practical as well as methodological. Solar-sail and drag-sail concepts are increasingly demonstrated in low Earth orbit on low-cost CubeSat and small-satellite platforms — NanoSail-D [16], CubeSail [17], InflateSail [18], and CanX-7 [19] — precisely because LEO offers a cost-effective, rapidly accessible test environment for maturing sail technology before committing to expensive deep-space missions [20, 21]. Reliable pre-flight prediction of the coupled SRP–drag dynamics for such demonstrators requires propagation tools whose force models have been verified against a recognised

reference. Open-source propagators (GMAT [27], Orekit [28], poliastro [29], Basilisk [30], Tudat [31]) make such studies reproducible and inspectable, but they must be validated; cross-checking an open-source propagator against STK/HPOP is an established verification practice [26, 27].

9.1. Open-source propagator

The ~~independent~~ propagator integrates the equations of motion of Section 4 directly in Cartesian ECI coordinates with a fixed-step fourth-order Runge–Kutta scheme at a production step of 10 s (chosen in Section 9.2). The force model comprises: (i) central gravity with the ~~J_2~~ J_2 zonal term; (ii) atmospheric drag ~~using the piecewise-exponential~~ with the frontal area $A|\cos\theta|$ projected onto the instantaneous inertial velocity, no atmospheric co-rotation, and the U.S. Standard Atmosphere 1976 density ~~profile [40] with a frontal area $A|\cos\theta|$ projected onto the instantaneous velocity and no atmosphere co-rotation~~[41] evaluated at the geocentric altitude and interpolated exponentially (log-linearly) between the tabulated values (Section 4); and (iii) solar radiation pressure ~~represented by a single body-fixed flat plate .~~ from a single one-sided flat plate with absorptivity $\alpha_a = 0.3$, specular reflectivity $\rho_s = 0.7$ and diffuse reflectivity $\rho_d = 0$, so that the normal-incidence coefficient is $C_r = 1.7$ (Table 2), with the solar flux scaled by $(1\,\mathrm{AU}/d_\odot)^2$ using the analytic Sun of Section 4. A binary cylindrical umbra gates the SRP. The sail normal is constructed once, at epoch, from the ~~local time of descending node~~LTAN, inclination and pitch angle and then frozen in inertial space, consistent with the passively spin-stabilised assumption. ~~A binary cylindrical umbra gates the SRP.~~ Mean (secular) elements are recovered by stroboscopic sampling of the osculating semi-major axis at ascending-node crossings, ~~which removes the ± 9 km J_2~~ the first sample falling one nodal period after epoch; this removes the ± 9 km J_2 short-period signature that would otherwise mask the few-kilometre secular signal of interest. A run stops when the perigee altitude falls below 120 km, which is reported as re-entry.

The SRP plate ~~is modelled with~~ uses the physically consistent specular/diffuse/absorptive decomposition of the flat-plate force [38, 39], which places the specular component along the plate normal with a ~~$\cos^2\alpha$ dependence~~ $\cos^2\alpha$ dependence, exactly the form of ~~the reduced mechanism model (~~Eq. ~~8). Calibrating the plate to specular reflectivity $\rho_s = 0.7$ and diffuse reflectivity $\rho_d = 0$ gives a normal-incidence coefficient~~

~~$C_{r,\mathrm{eff}} = 1 + \rho_s = 1.7$, matching the reflectivity coefficient of Table 2. An offline check confirms that this plate reproduces the paper's $C_r \cos^2 \alpha$ acceleration magnitude to within 1.6% for incidence angles $\alpha \leq 45°$.~~ (8); its agreement with the $C_r \cos^2 \alpha$ law is checked against Orekit's independent plate implementation in Section 9.5. We emphasise that the along-normal (specular) direction is essential: an SRP force of the same magnitude directed along the Sun line performs zero net orbit-averaged work in this geometry~~(~~, because the positive and negative arcs cancel~~)~~, so only the normal-directed plate reproduces the eclipse-gated pumping mechanism. Section 6 quantifies this at the gain anchor: $\Delta a(14\,\mathrm{d}) = +5.95\,\mathrm{km}$ with the plate, $-9.89\,\mathrm{km}$ without SRP, and $-10.50\,\mathrm{km}$ with the Sun-line variant of the force.

9.2. Step-size convergence

The umbra makes the SRP force discontinuous in time, and a fixed-step integrator that straddles the shadow boundary commits an error of first order in the step at every eclipse entry and exit. Figure 11 shows the consequence for the 14-day node-sampled $\Delta a(14\,\mathrm{d})$ at the 800 km gain anchor and at the former 700 km anchor for three configurations: the full model, the full model with the umbra disabled, and drag only. At 800 km the full model gives -1.9898, $+5.6508$, $+5.9329$, $+5.9521$, $+5.9508$, $+5.9546$, $+5.9538$ and $+5.9556$ km at $\Delta t = 120$, 60, 30, 15, 10, 7.5, 5 and 2.5 s, against $+5.9401$ km for an adaptive Dormand–Prince 8(5,3) (DP853) reference in Orekit with eclipse event detection, which places the integration steps exactly on the shadow boundaries. The two smooth configurations converge by 30 s: the umbra-disabled model gives -10.5934 km at 30 s and -10.5873 km at 10 s against the DP853 value of -10.5864 km, and drag only gives -9.9091 and -9.8938 km against -9.8989 km. Running the same RK4 ladder inside Orekit confirms that the behaviour belongs to the problem, not to the implementation: a least-squares fit gives an observed convergence order of 0.94 for the eclipse-gated model, against the nominal fourth order, while the smooth configurations are already converged to within about 0.01 km at 30 s so that no meaningful order can be fitted. The coarse-step error is also non-asymptotic: in the Orekit ladder the 60 s result (+7.229 km) overshoots the reference by more than the 120 s result (+6.068 km), and in the present propagator the 120 s step reverses the sign of the result while 60 s undershoots by 0.3 km. Accuracy therefore cannot be inferred from the step size alone for this problem; it must be established

against an event-detecting reference. The production step of 10 s reproduces the DP853 reference to +0.18% (+5.9508 against +5.9401 km); at 700 km the full model gives −21.0735 km at 10 s against −21.0640 km at 2.5 s.

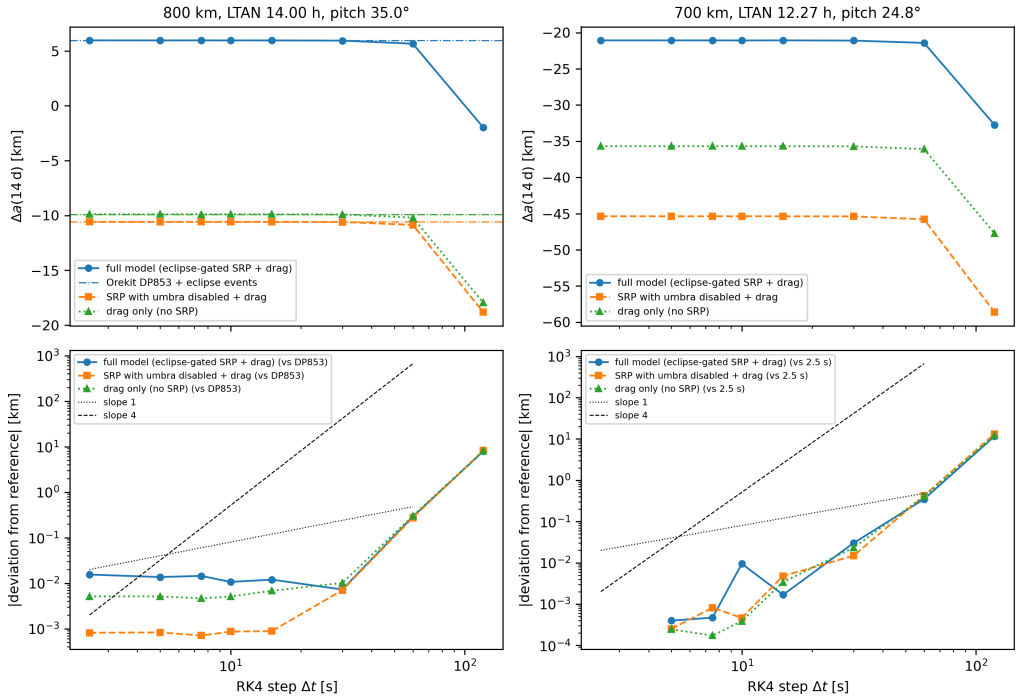


Figure 11: Step-size convergence of the node-sampled $\Delta a(14\,\mathrm{d})$ for fixed-step RK4 at $\Delta t = 120$ to 2.5 s, at the 800 km gain anchor (left; LTAN 14.0 h, pitch +35°) and the former 700 km anchor (right; LTAN 12.266 h, pitch +24.8°) for the full model, the full model with the umbra disabled, and drag only. Dash-dot lines are the Orekit DP853 references with eclipse event detection at 800 km: +5.9401, −10.5864 and −9.8989 km. Lower row: absolute deviation from the reference (the DP853 value at 800 km; the 2.5 s RK4 result at 700 km, where no Orekit reference was run) with first- and fourth-order slope guides. The smooth configurations converge by 30 s; the eclipse-gated model converges only slowly (observed order about one in the Orekit ladder) and settles within 0.02 km of the reference below 15 s; the production step of 10 s reproduces the reference to 0.18%.

9.3. STK Astrogator configuration

The STK reference cases were built ~~entirely through the STK 13 Object Model API driven from Python, running the STK Engine headlessly so that large parameter~~ with Ansys STK Engine 13.0.1 (Premium; installer

Ansys_STK_Premium_ODTK_v13.0.1), driven headlessly through the agi.stk13 Object Model from CPython 3.12 on Windows 11, so that the sweeps could be executed without the graphical interface. Each satellite uses the Astrogator propagator: a Mission Control Sequence with an Initial State segment (Keplerian elements in the Earth ~~inertial frame~~ICRF frame, epoch 1 Jan 2026 12:00 UTC) and a Propagate segment terminated on a ~~duration~~ Duration stopping condition. A ~~dedicated numerical propagator was cloned in the component browser and its force-function collection~~ numerical propagator cloned from "Earth Point Mass" carries three propagator functions configured to match the paper's model: ~~a gravity-field function~~ "Gravitational Force" truncated to degree 2, order 0 (~~J_2~~ J_2 only, tides ~~disabled); an atmospheric-drag function using the built-in U.S. Standard Atmosphere~~off); "US Standard Atmosphere" (1976~~density model ; and an~~) with the spherical area model for the attitude-independent validation and the N-plate ~~SRP function. The N-plate SRP and drag geometries are supplied through STK plate-definition files specifying, per plate , the area , body-frame normal, and (for SRP) specular and diffuse reflectivities; the sail is represented by a single body-fixed $+Z$ plate for SRP and by two opposed $\pm Z$ plates for drag~~ model otherwise; and "NPlate SRP" with the cylindrical shadow and a mean flux of 1367.2 W m^{-2}, equal to $P_{1\,\mathrm{AU}}c$ of the open-source plate. The plate files define one body $+Z$ SRP plate (area 100 m^2, specular reflectivity 0.70, diffuse 0.00) and two opposed $\pm Z$ drag plates ($C_D = 2.2$), so that the projected ~~area follows $A|\cos\theta|$. The inertially-fixed~~ drag area follows $A|\cos\theta|$. The inertially fixed attitude is imposed with the ~~standard~~ `eProfileFixedInAxes` profile ~~and a quaternion mapping the body $+Z$ axis~~ referenced to Earth ICRF and the quaternion mapping body $+Z$ onto the sail normal. Osculating elements are read from the Classical Elements data provider ~~and, where a secular quantity is required, orbit-averaged over one nodal period. The full build, sweep and post-processing pipeline is provided in the accompanying repository(Section 9.8)~~ every 60 s; for both tools the mean semi-major axis of this comparison is the centred one-period average of the osculating value, and the quoted change is taken between $t = T/2$ and $t = T/2 + 14$ d, so that it spans exactly 14 days. (The open-source values in Table 8 therefore differ slightly from the node-sampled values used elsewhere in the paper, e.g. +5.85 against +5.95 km at the gain anchor.)

A script-based force-plugin route was attempted first and then abandoned. The Python force-model plugins of STK 13 require a registered COM

server implementing the drag and SRP plugin interfaces, and registration failed on this installation. The attempt is documented in the repository, and the built-in N-plate models were used instead. Faceted and N-plate SRP models of this kind are the standard higher-fidelity representation in the flight-dynamics literature [43–46], but they are configured through several layers of objects whose effective values are not all visible through the properties one sets; the comparison below therefore reads STK's *effective* inputs back along the trajectory through the "Astrogator Values" data providers (applied density, drag area, SRP area, mass) rather than trusting the configured ones.

~~Figure 12 compares the two propagators for the conservative and dissipative parts of the model~~ That read-back exposed two configuration errors in the STK build of the originally submitted version, which must be stated plainly because they account for its two principal findings. (i) The Initial State segment's fuel tank carried its default 50 kg of fuel, which Astrogator adds to the 10 kg dry mass when evaluating drag and SRP. The submitted STK runs therefore propagated a 60 kg vehicle, $A/m = 1.67$ instead of $10\,\mathrm{m^2\,kg^{-1}}$, while every read-back property (dry mass, drag area, C_D, density) looked correct because the fuel mass is stored elsewhere. This is the whole of the "order-of-magnitude" drag discrepancy of the submitted version (-7.4 km for STK against -84 km for the open-source propagator at 700 km): with the tank zeroed, the 14-day spherical-drag decay changes from -7.48 to -60.30 km at ~~700 and 800 km~~700 km and from -2.77 to -17.68 km at 800 km (Table 8). (ii) The Object Model's `AssignQuaternion` method (interface `IAgOrientationQuaternion`) takes (q_x, q_y, q_z, q_s) with the scalar component last, and had been called scalar-first. The resulting attitude mis-oriented the plate: STK's effective Sun-facing area at epoch read $16.43\,\mathrm{m^2}$ where $A\cos\alpha = 82.90\,\mathrm{m^2}$ was expected (probe at pitch 34°), and the 3-day J_2+SRP response was $+0.21$ km against $+3.28$ km for the open-source propagator. This, not a limitation of the N-plate model, was the origin of the "N-plate model does not use the attitude" finding of the submitted version. With the Object-Model component order the effective area reads $82.96\,\mathrm{m^2}$ and the 3-day response $+3.29$ km; at the 35° gain anchor the effective area is $81.97\,\mathrm{m^2}$ ($A\cos 35° = 81.92\,\mathrm{m^2}$), and at the former 24.8° anchor $90.77\,\mathrm{m^2}$ ($90.78\,\mathrm{m^2}$). STK's SRP response then matches the open-source propagator, as the next subsection shows.

9.4. Cross-validation against STK

Figure 12 and Table 8 compare the two tools configuration by configuration. For the conservative dynamics (two-body plus ~~J_2, SRP and drag~~ J_2, drag and SRP disabled) the centred-mean semi-major ~~axis of the two tools agrees to better than 0.05 km over a 14-day propagation, confirming that~~ axes are identical to the metre level: the maximum absolute difference over the 14-day window is 0.25 m at 700 km and 0.16 m at 800 km, confirming the gravitational model, the coordinate handling and the mean-element extraction of the open-source propagator~~are correct~~.

~~The atmospheric-drag comparison is more nuanced and constitutes the principal validation finding. At a matched state the *instantaneous* dragacceleration computed from STK's reported density, drag area, drag coefficient and mass agrees with the~~ For J_2 plus spherical drag, with the fuel tank corrected, STK gives -60.295 km against -70.564 km for the open-source ~~value to within $\sim$4% (both $\approx 1.9 \times 10^{-5}\,\mathrm{m\,s^{-2}}$ at 700 km, with identical density $\approx 3.1 \times 10^{-14}\,\mathrm{kg\,m^{-3}}$). Nonetheless, the *propagated* 14-day mean semi-major-axis decay differs by close to an order of magnitude (-84 km open-source versus -7 km STK at 700 km). Because the density, area, coefficient and instantaneous force all match, the divergence arises in how each tool applies drag over the trajectory — plausibly the atmosphere-relative velocity~~ propagator at 700 km (ratio 0.854) and -17.678 against -18.547 km at 800 km (ratio 0.953); with the N-plate drag model at the gain-anchor attitude the corresponding values are -27.972 against -31.524 km (ratio 0.887) at 700 km and -9.446 against -10.015 km (ratio 0.943) at 800 km. The as-submitted default-tank values of -7.482 and -2.770 km are plotted dotted in panel (b) and listed in the table for the record. The residual 5–15% is a difference of convention, not of force magnitude. Astrogator applies a co-rotating atmosphere and evaluates the density at the geodetic altitude, whereas the paper's model uses the inertial velocity and the geocentric altitude. Orekit, switching these one at a time with everything else fixed (Section 9.6), gives factors of $\times 1.034$ (~~co-rotation)treatment, the geodetic-versus-geocentric altitudeused for the density lookup,~~ co-rotation) and $\times 0.816$ (geodetic altitude) at 700 km, $\times 1.024$ and ~~differences in each tool's secular-element definition (STK's Brouwer–Lyddane mean elements, derived from conservative theory, are known to be unreliable under non-conservative forcing). This is exactly the kind of discrepancy that validation is meant to expose: for the~~ $\times 0.901$

at 800 km, and ×0.842 and ×0.922 combined, which accounts for the STK/open-source ratios to within about 1.5% at 700 km and 3% at 800 km. The density STK actually applied, read back along the trajectory, averages 0.887 (range 0.757–1.032, over 632–699 km) and 0.937 (range 0.839–1.040, over 774–799 km) of the open-source value at the same geocentric altitude, consistent with the geodetic-altitude evaluation. For the extreme area-to-mass ratio of a sail (~~$A/m = 10\,\mathrm{m^2\,kg^{-1}}$) ,~~ $A/m = 10\,\mathrm{m^2\,kg^{-1}}$) atmospheric drag is the dominant and ~~least-portable~~ least portable element of the force model [47], and ~~its treatment~~ these conventions must be stated ~~explicitly~~ for any LEO sail prediction to be reproducible; they are now stated in Section 4.

~~The SRP comparison surfaced a further practical limitation. In our configuration STKAstrogator's built-in~~ With the quaternion corrected, STK's N-plate SRP ~~model did not apply the satellite's inertially-fixed attitude to the plate body frame:~~ response at the gain anchor agrees with the open-source plate: +15.554 against +15.565 km over 14 days with eclipse on (maximum difference 0.014 km, about 0.1%), and +0.004 against +0.006 km with the umbra removed, so both tools reproduce the cancellation of the orbit-averaged SRP work under continuous illumination that Section 3 predicts. For the full model at the gain anchor STK gives +6.441 km with eclipse on and −10.134 km with it off, against +5.846 and −10.927 km for the ~~effective Sun-facing plate area reported by STK was zero for the passively oriented sail, so the model produced no SRP force for this attitude mode. Industry flight-dynamics N-plate models are designed principally for actively pointed or Sun-tracking spacecraft,~~ open-source propagator: the eclipse-driven sign reversal of Section 6 is reproduced, and STK's slightly larger gain follows directly from its 5% weaker drag. Over 75 days STK reproduces the transient signature of Section 7: its mean semi-major axis peaks at +12.92 km on day 41.9 and ends at +2.56 km, against a peak of +10.60 km on day 38.0 and ~~are not readily configured for a plate whose normal is frozen in inertial space. We therefore validated the~~ −2.70 km at day 75 for the open-source ~~SRP implementation analytically, as described above, rather than against STK~~propagator under the centred-mean definition (+10.86 km on day 38.1 and −2.69 km node-sampled, Table 5); the end-point difference of 5.3 km is the drag-convention difference accumulated over 1069 orbits of decay. At the former 700 km anchor the full model decays in both tools, −17.357 km in STK against −21.548 km. ~~This limitation is itself a useful result: it shows that verifying passive-sail SRP dynamics~~

~~benefits from a purpose-built, inspectable propagator,~~

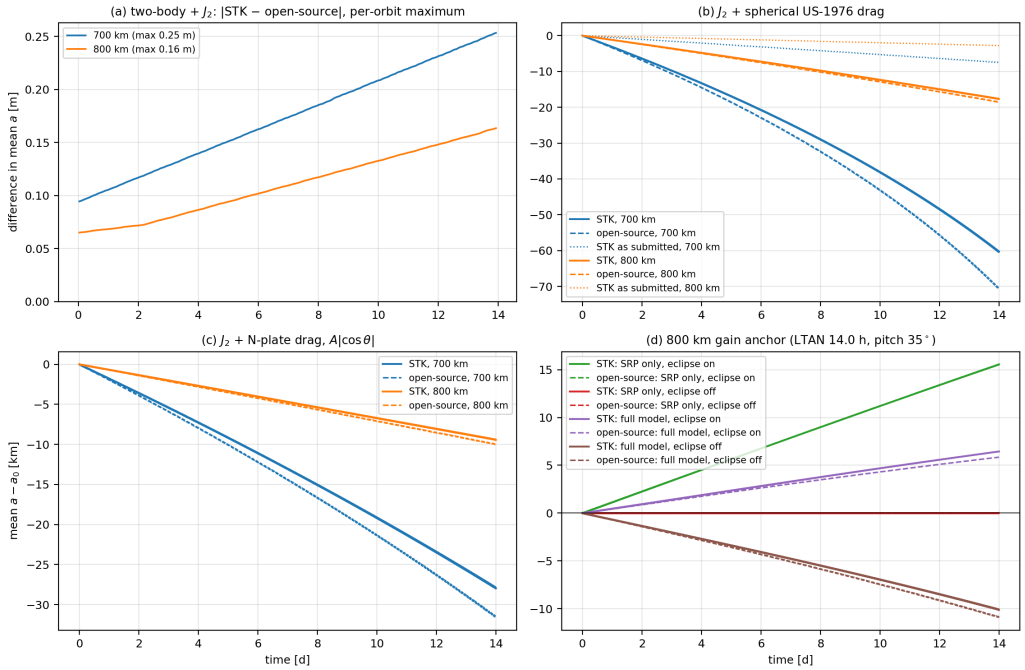


Figure 12: Cross-validation of the open-source propagator against STK Astrogator (STK Engine 13.0.1), centred one-period mean semi-major axis over 14 days. (a) Two-body plus J_2 at 700 and 800 km: per-orbit maximum of the absolute difference between the two tools, which stays below 0.25 m over 14 days. (b) J_2 plus spherical U.S. Standard Atmosphere 1976 drag with the fuel tank corrected (solid) and the as-submitted default-tank curves (dotted). (c) J_2 plus N-plate drag at the gain-anchor attitude. (d) SRP only and the full model at the 800 km gain anchor with eclipse on and off; the sign reversal under eclipse is reproduced by both tools.

9.5. Cross-validation against Orekit

STK validates the conservative dynamics, the application of drag and the SRP response, but its force models are closed. The second reference is therefore Orekit 13.1.7 (Hipparchus 4.0.3), driven through `orekit_jpype`, whose force models, attitude handling, planetary ephemerides and adaptive integrators are independent of the present code and inspectable. The mapping is one-to-one: `NewtonianAttraction` and ~~it is consistent with the higher-fidelity SRP modelling literature, in which faceted and N-plate modelsare tailored to specific spacecraft~~

Table 8: Cross-validation against STK Astrogator: change in the centred one-period mean semi-major axis over 14 days (75 days for the last two rows), STK against the open-source propagator (OS), the ratio STK/OS, and the maximum absolute difference max$|\Delta|$ between the two over the window; all changes in km. Gain anchor LTAN 14.0 h, pitch $+35°$; the former-anchor row uses LTAN 12.266 h, pitch $+24.8°$. The "default tank" rows are the build of the original submission (60 kg vehicle), shown for the record.

Configuration	Alt. [km]	STK	OS	STK/OS	max$\|\Delta\|$
Two-body + J_2	700	-0.055	-0.055	0.999	0.0003
Two-body + J_2	800	-0.000	-0.000	—	0.0002
J_2 + spherical drag, default tank	700	-7.482	-70.564	0.106	63.196
J_2 + spherical drag, default tank	800	-2.770	-18.547	0.149	15.817
J_2 + spherical drag, corrected	700	-60.295	-70.564	0.854	10.535
J_2 + spherical drag, corrected	800	-17.678	-18.547	0.953	0.885
J_2 + N-plate drag	700	-27.972	-31.524	0.887	3.678
J_2 + N-plate drag	800	-9.446	-10.015	0.943	0.571
J_2 + N-plate SRP, eclipse on	800	$+15.554$	$+15.565$	0.999	0.014
J_2 + N-plate SRP, eclipse off	800	$+0.004$	$+0.006$	0.758	0.002
Full model, eclipse on	800	$+6.441$	$+5.846$	1.102	0.597
Full model, eclipse off	800	-10.134	-10.927	0.927	0.796
Full model, eclipse on, former anchor	700	-17.357	-21.548	0.805	4.193
Full model, 75 d, peak	800	$+12.922$	$+10.604$	—	—
Full model, 75 d, final	800	$+2.560$	-2.695	—	5.275

~~geometries and pointing modes [43–46].~~ `J2OnlyPerturbation` with the paper's constants; a `FrameAlignedProvider` for the frozen sail normal; a `BoxAndSolarArraySpacecraft` with one one-sided `FixedPanel` (absorption 0.3, reflection 0.7) for SRP through `RadiationPressureModel` with a `CylindricallyShadowedLightFluxModel`, and the same panel double-sided in a separate spacecraft for `DragForce`; `IsotropicDrag` for the spherical validation case; the paper's US-1976 interpolation injected as an Orekit `Atmosphere`, or Orekit's own Harris–Priester and NRLMSISE-00 models; and the Sun either from the paper's analytic series, to isolate the propagators, or from the JPL DE-440 ephemeris. Like-for-like comparisons use fixed-step RK4 at 30 s in both tools; the reference runs use DP853 with eclipse event detection. Table 9 collects the results.

~~Using the validated~~ At the level of the force models, the two implementations are the same to round-off. At 198 states along one orbit the maximum relative difference of the accelerations is 0 for gravity, 7.4×10^{-15} for drag and 1.1×10^{-15} for SRP; the eclipse flag agrees at 198 of 198 states; the plate normal recovered from Orekit's propagated attitude matches the paper's construction to 2.8×10^{-17}, and the plate force follows the $\cos^2 \alpha$ law to 1.5×10^{-15}. The propagated 14-day mean semi-major axis (RK4 30 s, spherical drag, no SRP) is then identical to within 4×10^{-10} km: -0.01198 km (700 km) and -0.01128 km (800 km) with J_2 only, -69.872 and -18.382 km with J_2 plus drag, in both tools. The SRP attitude coupling that the submitted STK build could not exercise is confirmed by Orekit's N-plate: the effective Sun-facing area at the former 24.8° anchor is 90.76 m^2 against the analytic 90.78 m^2, the plate acceleration is 80.2 μm s^{-2} at normal incidence and zero at $\alpha = 90°$, and over a sweep of incidence angles the N-plate magnitude departs from $C_r \cos^2 \alpha$ by at most 1.55% for $\alpha \leq 45°$.

For the full model at the gain anchor the propagators agree, and the remaining difference is attributable to the Sun ephemeris. With the same analytic Sun held at 1 AU in both tools (RK4 30 s), the open-source propagator ~~we re-ran the parameter-space sweep and the representative cases. The existence of the passive-gain regime is confirmed independently:~~ gives $\Delta a(14\,\mathrm{d}) = +5.438$ km and Orekit $+5.605$ km, a maximum difference along the window of 0.181 km; with each tool's production Sun — the analytic series with flux $\propto 1/d_\odot^2$ here, DE-440 in Orekit — the values are $+5.933$ and $+6.127$ km (maximum difference 0.194 km). The Sun ephemeris alone (analytic against DE-440, everything else fixed) moves the 14-day gain by

0.522 km, about 9%, a sensitivity worth stating: the epoch lies within days of perihelion, the analytic direction departs from DE-440 by up to 0.369° over 75 days, and the Sun distance departs from 1 AU by up to 1.67%. At the former 700 km anchor the corresponding values are −21.478 against −21.180 km and −21.158 against −20.842 km, both decaying.

Figure 13 ~~reproduces a single connected lobe of positive $\Delta a_{\max}$ over a bounded band of LTDN and *positive* pitch angle, in agreement with the location of the lobe in Figure 2. The~~ compares the 800 km map of $\Delta a_{\max,14\mathrm{d}}$ with its Orekit counterpart, computed with DP853 and eclipse event detection on a 24×19 subset of the 48×37 grid of Figure 2 (LTAN step 1 h, pitch step 10°; 456 common cells, none failed). Both maps have 82 positive cells, the sign agrees in every cell, and both peak at (14.0 h, +30°), the nearest common cell to the +35° anchor, with 5.776 km here and 5.753 km in Orekit. The cell-by-cell difference has a mean of 0.003 km, an RMS of 0.029 km (0.068 km over the positive cells) and a maximum of 0.236 km. The eclipse-causality ~~result is likewise reproduced : toggling the umbra on and off at a representative gain geometry flips the sign of the short-horizon semi-major-axis response (from growth to decay), confirming that eclipse timing — not merely SRP magnitude — controls the sign of the orbit-averaged SRP work. The~~ test of Section 6 is reproduced over 30 days at the gain anchor: Orekit gives +10.33 km with eclipse on, still rising at day 30, and −28.53 km with it off, while the former 700 km anchor gives −90.66 and −459.0 km. Over 75 days the Orekit gain case peaks at +10.95 km on day 37.9 and ends at −2.37 km with $e = 0.01225$, apogee 885.6 km and perigee 709.7 km, against +10.86 km on day 38.1, −2.69 km, $e = 0.0122$, 884.7 and 709.9 km in Table 6; the Orekit propagation of the former 700 km anchor ends on day 39.3 at −412.5 km with the perigee at 275 km, the counterpart of the re-entry on day 39.5 in Table 5. The one-orbit ~~SRP power profile (Figure 16) shows the eclipse removing the negative-work arc,~~ SRP work budget of Figures 4 and 5 is reproduced as well: Orekit's umbra occupies 32.0% of the gain-anchor orbit (32.1% here), 77.6% of the eclipsed samples carry negative no-eclipse SRP power (77.4%), and the cumulative SRP work over the orbit is +309.7 J kg^{-1} with eclipse on and +0.77 J kg^{-1} with it off (+310.3 and ~~the long-horizon evolution (Figure 17) reproduces the characteristic transient: an early apogee-pumping rise in mean semi-major axis followed by a self-limiting decline driven by eccentricity growth and perigee erosion.~~ +1.4 J kg^{-1}).

~~The independent reproduction also sharpens the interpretation of the effect. In the~~ In summary, Orekit validates the drag law, the SRP attitude coupling, the eclipse gating and the gain regime itself independently of the present implementation, and STK validates the conservative dynamics, the application of drag and, with the corrected configuration, the SRP response. The regime is a property of the dynamics, not of one code.

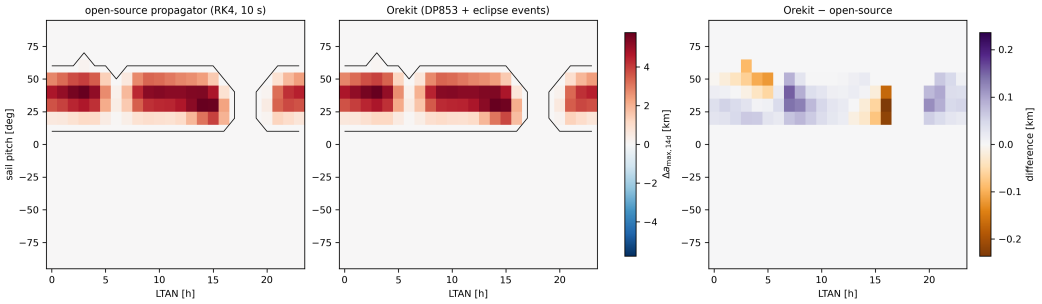


Figure 13: Independent reproduction of the 800 km gain lobe. Left: $\Delta a_{\max,14\mathrm{d}}$ from the open-source propagator on the 24 × 19 cells common to the Orekit sweep (LTAN step 1 h, pitch step 10°). Centre: the same map from Orekit (DP853 with eclipse event detection). Right: the cell-by-cell difference (Orekit minus open-source). Both maps contain 82 positive cells, agree in sign everywhere and peak at LTAN 14.0 h, pitch +30° (5.776 and 5.753 km); the RMS difference is 0.029 km and the largest 0.236 km.

9.6. Drag conventions and atmosphere models

The STK comparison leaves drag as the one component whose propagated effect differs between tools, and Section 8 shows that the regime boundary at 800 km lies within a factor of two of the US-1976 density. Table 10 therefore separates the drag conventions from the atmosphere model with Orekit, propagating J_2 plus spherical drag for 14 days (RK4 30 s, node-sampled) and changing one element at a time. At 700 km the baseline (US-1976, geocentric altitude, no co-rotation) gives −69.87 km; a rigidly co-rotating atmosphere gives ×1.034, the geodetic altitude ×0.816, and the two together ×0.842. At 800 km the baseline is −18.38 km and the factors are ×1.024, ×0.901 and ×0.922. The conventions are thus worth 8–16%, and they account for the STK residual of Section 9.4 to within a few per cent. The atmosphere model is worth far more. Orekit's built-in Harris–Priester model multiplies the decay by 7.08 at 700 km, where the sail re-enters on day 10.45, and by 3.15 at 800 km; NRLMSISE-00 driven by the CSSI space-weather file for January

Table 9: Cross-validation against Orekit 13.1.7. Force gates are evaluated at 198 states along one orbit with a shared Sun position; propagated 14-day values use RK4 at 30 s in both tools and the node-sampled mean semi-major axis; the map, 75-day and one-orbit rows use Orekit's DP853 integrator with eclipse event detection. Anchor LTAN 14.0 h, pitch +35° unless stated.

Quantity	Open-source	Orekit	Difference
Gravity acceleration, relative	—	—	0
Drag acceleration, relative	—	—	7.4×10^{-15}
SRP acceleration, relative	—	—	1.1×10^{-15}
Eclipse flag	—	—	0 mismatches
Sail normal from attitude	—	—	2.8×10^{-17}
Plate force vs $\cos^2 \alpha$ law	—	—	1.5×10^{-15}
J_2 only, 14 d, 700 km [km]	−0.01198	−0.01198	2.6×10^{-10}
J_2 only, 14 d, 800 km [km]	−0.01128	−0.01128	2.7×10^{-10}
J_2 + spherical drag, 14 d, 700 km [km]	−69.872	−69.872	4.0×10^{-10}
J_2 + spherical drag, 14 d, 800 km [km]	−18.382	−18.382	2.8×10^{-10}
Sun-facing area at pitch 24.8° [m^2]	90.78 (analytic)	90.76	0.02
N-plate magnitude vs $C_r \cos^2 \alpha$, $\alpha \leq 45°$	—	—	$\leq 1.55\%$
Full model, analytic Sun at 1 AU [km]	+5.438	+5.605	0.181 (max)
Full model, production Sun ($1/d_\odot^2$ / DE-440) [km]	+5.933	+6.127	0.194 (max)
Former 700 km anchor, analytic Sun [km]	−21.478	−21.180	0.299 (max)
Former 700 km anchor, production Sun [km]	−21.158	−20.842	0.318 (max)
800 km map, 456 common cells: positive cells	82	82	sign agreement 100%
peak at (14.0 h, +30°) [km]	5.776	5.753	RMS 0.029, max 0.236
75 d: peak [km] (day)	+10.86 (38.1)	+10.95 (37.9)	0.09
75 d: final [km]	−2.69	−2.37	0.32
One orbit: umbra fraction	32.1%	32.0%	0.1%
One orbit: sign-selectivity	77.4%	77.6%	0.2%
One orbit: SRP work, eclipse on [J kg^{-1}]	+310.3	+309.7	0.6
One orbit: SRP work, eclipse off [J kg^{-1}]	+1.4	+0.77	0.6

2026 multiplies it by 2.57 and 1.59. Because Orekit's built-in models are evaluated with their own conventions (geodetic altitude and co-rotation), these factors include the ×0.84–0.92 convention effect, and the model effect proper is slightly larger.

A propagator-free check settles the question of scale raised by the reviewers. Integrating the secular rate $\mathrm{d}a/\mathrm{d}t = -\rho\, C_D\, (A/m)\sqrt{\mu a}$ with the same density table along the decaying orbit gives −63.9 km at 700 km and −17.6 km at 800 km, within 8.5% and 4.2% of the propagated −69.87 and −18.38 km; the remainder is the growth of eccentricity and the node-sampling convention. Holding the density at its 700 km value instead gives about −43 km, which is the reviewers' estimate: the difference is the density increase along the 60–70 km of decay, not a defect of either tool. The three figures of the original submission are therefore reconciled as follows: −43 km is the constant-density estimate, −84 km was the value of the open-source ~~model the clearest gain lobe appears at 800 km, whereas at 700 km the same geometry is drag-marginal; the peak reproduced excursion (∼+7 km over the sampled window)is smaller than the reduced-model estimate.~~ propagator before the correction of the atmosphere table (Section 4), and −7 km was STK propagating a 60 kg vehicle; the corrected values are −70 km (open-source, Orekit) and −60 km (STK, with its own conventions). At this altitude drag wins for every geometry (Figure 2) and the former anchor re-enters on day 39.5 (Table 5).

The ~~density-sensitivity sweep makes the cause explicit: reducing the atmospheric density recovers multi-kilometre gain even at 700 km, so the altitude~~ US-1976 profile is a static reference that carries no dependence on solar or geomagnetic activity (Section 4). To locate the regime boundary under a realistic thermosphere, the full model at the gain anchor was propagated for 14 days with the US-1976 baseline and with NRLMSISE-00 at constant low, moderate and high activity ($F_{10.7} = 70$, 140 and ~~magnitude at which the regime becomes net-positive are governed primarily by the assumed drag model. Read together with the order-of-magnitude drag discrepancy between~~ 200; $A_p = 4$, 15 and 40) and with the CSSI space-weather file, at 800 and 700 km; Figure 14 and Table 11 report $\Delta a_{\max,14\mathrm{d}}$ and $\Delta a(14\,\mathrm{d})$.

At the 800 km gain anchor the regime survives every atmosphere except the high-activity one. Under low activity ($F_{10.7} = 70$, $A_p = 4$) the 14-day

Table 10: Drag conventions and atmosphere models in Orekit: 14-day change of the node-sampled mean semi-major axis for J_2 plus spherical drag ($A/m = 10\,\mathrm{m^2\,kg^{-1}}$, $C_D = 2.2$, RK4 at 30 s), with the factor relative to the baseline of the paper (US-1976, geocentric altitude, no co-rotation). Orekit's built-in models use their own conventions (geodetic altitude, co-rotation); NRLMSISE-00 is driven by the CSSI space-weather file for January 2026. With Harris–Priester the 700 km sail re-enters on day 10.45. The last row is the propagator-free secular integral with the same density table.

	700 km		800 km	
Configuration	$\Delta a(14\,\mathrm{d})$ [km]	factor	$\Delta a(14\,\mathrm{d})$ [km]	factor
Baseline (conventions of the paper)	−69.87	1.000	−18.38	1.000
Co-rotating atmosphere	−72.28	1.034	−18.82	1.024
Geodetic altitude	−57.04	0.816	−16.57	0.901
Co-rotating + geodetic	−58.82	0.842	−16.95	0.922
Harris–Priester (Orekit)	−494.7†	7.08	−57.84	3.15
NRLMSISE-00 (Orekit)	−179.9	2.57	−29.17	1.59
Analytic secular integral	−63.9	0.915	−17.6	0.958

†At re-entry, day 10.45.

gain rises to +13.18 km, against +5.94 km with US-1976; under moderate activity (140, 15) it falls to +2.22 km, with the CSSI space-weather file for January 2026 to +0.50 km, and under high activity (200, 40) there is no gain at all ($\Delta a_{\mathrm{max,14d}} = 0$, $\Delta a(14\,\mathrm{d}) = -45.5$ km). The drag-only decays of the same runs, −2.33, −9.90, −12.72, −14.43 and −59.58 km respectively, show the cause: at this altitude and geometry the ~~two tools, this identifies atmospheric-drag modelling — not the SRP mechanism — as~~ low-activity NRLMSISE-00 drag is about a quarter of the US-1976 value, the moderate level and the January 2026 file about 1.3 and 1.5 times it, and the high level about six times. At 700 km the same geometry gains only under low activity (+10.69 km; drag only −4.69 km); it decays with US-1976 (−13.77 km; −31.15 km drag only), under moderate activity (−30.87; −45.15 km) and with the CSSI file (−38.52; −52.78 km), and under high activity it re-enters within the 14 days both with and without SRP.

The regime boundary therefore moves with the thermosphere. At 800 km the gain survives under low and moderate activity, is marginal under the January 2026 conditions and closes under high activity; at 700 km it exists only under low activity, where the quiet-Sun thermosphere is thinner than the static US-1976 profile. The US-1976 baseline of this paper thus lies between

the low and moderate NRLMSISE-00 levels and slightly below the January 2026 conditions: it is neither the most favourable nor the least favourable case. The NRLMSISE-00 runs use Orekit's implementation with its own conventions (geodetic altitude, co-rotating atmosphere), and all runs use the DP853 integrator with eclipse event detection, so the US-1976 row of Table 11 is the DP853 reference of Section 9.2 (+5.94 km) rather than the RK4 value of the maps.

Table 11: Space-weather dependence of the gain regime (Orekit, DP853 with eclipse event detection, 14 days, node-sampled mean semi-major axis): the full model and drag only at the gain-anchor geometry (LTAN 14.0 h, pitch +35°) at 800 and 700 km with the US-1976 baseline (conventions of the paper) and with Orekit's NRLMSISE-00 (geodetic altitude, co-rotation) at constant low, moderate and high activity (bracketed values are $F_{10.7}$ and A_p) and with the CSSI space-weather file for January 2026. All values in km; for the full model $\Delta a_{\max,14\,\mathrm{d}}$ / $\Delta a(14\,\mathrm{d})$, for drag only $\Delta a(14\,\mathrm{d})$. "Re-entry" marks runs whose perigee fell below 120 km within the 14 days.

Atmosphere model (activity)	800 km		700 km	
	full model	drag only	full model	drag only
	$\Delta a_{\max,14\,\mathrm{d}}$ / $\Delta a(14\,\mathrm{d})$	$\Delta a(14\,\mathrm{d})$	$\Delta a_{\max,14\,\mathrm{d}}$ / $\Delta a(14\,\mathrm{d})$	$\Delta a(14\,\mathrm{d})$
US-1976 (baseline)	+5.940 / +5.940	−9.899	0 / −13.766	−31.152
NRLMSISE-00, low (70, 4)	+13.178 / +13.178	−2.327	+10.685 / +10.685	−4.685
NRLMSISE-00, moderate (140, 15)	+2.215 / +2.213	−12.724	0 / −30.872	−45.154
NRLMSISE-00, high (200, 40)	0 / −45.455	−59.583	re-entry	re-entry
NRLMSISE-00, CSSI (Jan 2026)	+0.500 / +0.499	−14.426	0 / −38.517	−52.777

These results are consistent with the density-multiplier sweep of Section 8, which places the drag-limited boundary at the gain anchor between 1× and 2× the US-1976 density at 800 km and at or below 0.5× at 700 km. Judged by its drag-only decay, the ~~dominant uncertainty in predicting where the passive-gain regime is realisable~~low-activity atmosphere is equivalent to about a quarter of the US-1976 profile at 800 km and gives +13.18 km, close to the 13.24 km of the 0.25× sweep point; the moderate level and the January 2026 file, at about 1.3× and 1.5×, fall inside the band and still gain, by 2.22 and 0.50 km, so that the boundary at this geometry lies close

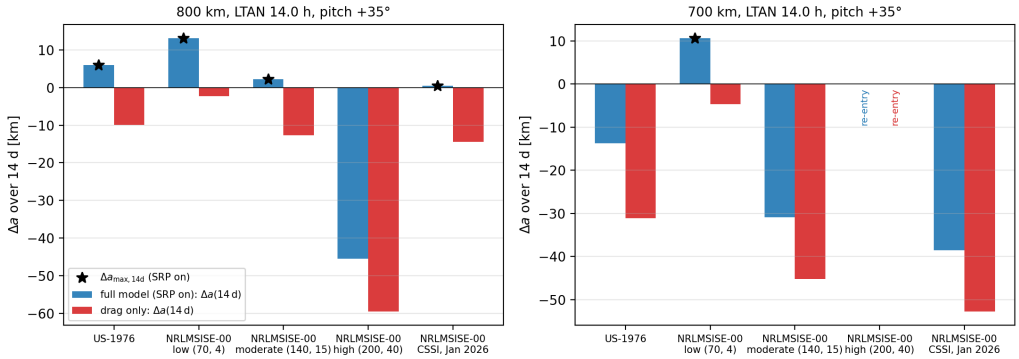


Figure 14: Space-weather dependence of the 14-day response at the gain-anchor geometry (LTAN 14.0 h, pitch +35°) at 800 km (left) and 700 km (right), from Orekit with DP853 and eclipse event detection. For each atmosphere model — the US-1976 baseline and NRLMSISE-00 at constant low ($F_{10.7} = 70$, $A_p = 4$), moderate (140, 15) and high (200, 40) activity and with the CSSI space-weather file for January 2026 — the bars give $\Delta a(14\,\mathrm{d})$ of the node-sampled mean semi-major axis for the full model (blue) and for drag only (red), and the stars mark $\Delta a_{\max,14\mathrm{d}}$ of the full model where it is positive. The 700 km runs under high activity re-enter within the 14 days and are marked "re-entry". The gain survives low and moderate activity at 800 km, is marginal under the January 2026 conditions and closes under high activity; at 700 km it exists only under low activity.

to 1.5× the US-1976 density; high activity, at about 6×, lies far beyond it, as does the Harris–Priester factor of 3.15 in Table 10. At 700 km the low-activity atmosphere is equivalent to about 0.15× the US-1976 profile, below the 0.5× threshold of the sweep, which is why a gain of +10.69 km appears there and nowhere else at that altitude. The mechanism ~~itself (eclipse-gated, geometry-selective, transient and self-limiting) is robust across both propagators; its quantitative threshold is drag-model dependent and should~~ of Sections 3 and 6 is independent of the atmosphere model; the altitude and activity level at which it becomes net-positive are not, and the quantitative boundary must always be reported together with the atmosphere model ~~used~~and the activity level assumed.

9.7. Note on the original submission

The quantitative results of Sections 6–8 in the originally submitted version of this paper were produced by a separate implementation of the same model, which was found during revision to contain errors in its aerodynamic incidence-angle convention, in its SRP coefficient and in its solar ephemeris. Those results have been withdrawn. Every result in the present version comes from the open-source propagator of Section 9.1, after the corrections described in this section, including the correction of the scale heights of the atmosphere table, which is now interpolated exponentially between the tabulated US-1976 values (Section 4). The two STK configuration errors of Section 9.3 were found by the same read-back procedure, and the gain regime is now realised at 800 km rather than at 700 km, where the corrected model is drag-dominated for every geometry.

9.8. Data and software availability

The open-source propagator, the STK Astrogator build~~and sweep~~, probe and read-back scripts, the Orekit cross-validation study with its force gates and atmosphere experiments, the plate-definition files, the ~~cross-validation utilities and all reproduced figures~~ result files (JSON and NPZ) behind every number in this paper and the scripts that generate every figure and table are archived in the project repository [48], so that the results of this section can be independently reproduced and extended.

10. Conclusions

This paper ~~examined~~ asked whether a passively oriented, spin-stabilised solar sail in near-polar low Earth orbit can exhibit a bounded interval of net

orbital-energy gain under eclipse-modulated solar-radiation-pressure forcing. The results show that such a regime does exist, but only over a ~~narrow and~~ well-defined region of the sampled ~~LTDN–pitch space~~ LTAN–pitch space and only at the higher of the two altitudes examined. At 800 km, 346 of the 1776 LTAN–pitch cells (19.5%) show a positive 14-day excursion of the mean semi-major axis; they lie exclusively at positive pitch, between +10° and +60° with the ridge at 35° to 40°, form one connected LTAN band from 20 h through midnight to 16 h, and peak at $\Delta a_{\mathrm{max,14d}} = 5.95$ km at LTAN 14.0 h, pitch +35°. At 700 km, with the same sail and atmosphere, none of the 1776 cells gains: the geometry that anchored the original submission decays by 21.07 km in 14 days and re-enters on day 39.5. The gain response is therefore not a generic property of passive sailing in LEO, but a constrained transient effect that appears only under specific Sun-relative geometries ~~.—~~and only where drag is weak enough to leave the SRP surplus intact.

The results also show that eclipse is the causal enabler of this ~~transient gain~~ regime. In the representative gain case ~~,—~~the inclusion of eclipse produces ~~positive short-horizon~~ +5.95 km of mean semi-major-axis growth by day 14, whereas removing ~~eclipse~~ the umbra from the same initial condition ~~eliminates that behaviour and leads instead to clear decay . Orbit-resolved~~ gives −10.59 km, within 0.7 km of the drag-only decay of −9.89 km; under continuous illumination the orbit-averaged SRP work of a fixed-attitude sail cancels, as the symmetry argument of Section 3 requires. A force of the same magnitude directed along the Sun line rather than the plate normal produces no gain either (−10.50 km), so the along-normal component carries the effect. The orbit-resolved SRP work diagnostics show why: eclipse does not merely weaken the SRP forcing, but selectively removes ~~unfavourable portions~~ one branch of the SRP ~~cycle. Under~~ power cycle, so that a cycle which cancels under continuous illumination (+1.4 J kg^{-1} per orbit) closes at +310.3 J kg^{-1} in the gain geometry ~~, this shifts the orbit-averaged SRP work balance positive; under neutral and decaying geometries, the same shadowing instead suppresses the positive recovery branch and leaves the net SRP contribution negative~~and at −315.4 J kg^{-1} in the decay geometry. The central physical result is therefore that eclipse timing ~~can control the sign~~controls the sign, and not merely the magnitude, of the orbit-averaged SRP work input.

At the same time, the longer-horizon behaviour shows unambiguously that ~~this~~ the effect is transient and self-limiting rather than sustained. The repre-

sentative gain trajectory ~~initially exhibits semi-major-axis increase, but this does not correspond to~~ rises to $\Delta a_{\max,75\mathrm{d}} = +10.86\,\mathrm{km}$ on day 38.1, returns to its initial semi-major axis on day 71.0 and ends the 75-day window at $-2.69\,\mathrm{km}$, with a cumulative total work of $-9.9\,\mathrm{kJ\,kg^{-1}}$. The early increase is not uniform orbit raising~~. Instead, the orbit evolves through apogee growth, perigee erosion, and rising eccentricity , so that~~ : apogee is raised by 111 km while perigee is lowered by about 90 km and the eccentricity grows to 0.0141, so the early energy advantage progressively creates the conditions that undermine it. ~~Drag becomes increasingly concentrated near perigee , while the favourable SRP projection weakens as the orbit distorts. As a result, the late-time~~ Two breakdown pathways act together. The first is perigee erosion, which concentrates the drag loss where the atmosphere is densest. The second, and here the dominant one, is attitude: the sail normal is fixed in inertial space while the Sun advances along the ecliptic and the Sun-synchronous plane precesses with it, so the Sun–normal angle opens from 35.0° to 83.4° and the SRP acceleration collapses by a factor of about fifty. The orbit-averaged energy balance turns negative ~~and the semi-major axis ultimately declines below its initial value~~at the peak of a, and everything after that point is decay. The transient gain regime ~~identified here~~ must therefore be understood as a front-loaded energy imbalance, not as evidence of persistent passive orbital uplift.

The sensitivity analysis supports the same interpretation. The ~~gain-anchor case~~ 800 km gain anchor is not a numerically isolated point~~, but neither~~ : it sits at the crest of a roll response at least 79° wide (half of the nominal response is retained from the −45° sampling limit to roughly +34°), and the gain rises monotonically, though with diminishing returns, with area-to-mass ratio, from 1.58 km at $2.5\,\mathrm{m^2\,kg^{-1}}$ to 10.59 km at $22.5\,\mathrm{m^2\,kg^{-1}}$. Neither is it broadly robust. ~~At lower altitude, the response is strongly limited by density and attitude offset; at higher altitude, the same geometry remains more tolerant over the sampled envelope, but still does not become a general passive orbit-raising solution~~The regime is bounded by drag: scaling the US-1976 density gives 13.24 km at 0.25×, 10.86 km at 0.5× and 5.95 km at 1×, but no gain at all at 2×, so the drag-limited boundary at 800 km lies between one and two times the reference profile. At 700 km the same geometry gains for no roll offset and no sail size in the sampled ranges, and only when the density is at most half the US-1976 value. The mechanism is therefore physically real and interpretable, yet intrinsically bounded by

geometry, ~~atmospheric conditions, and~~ by the atmospheric density, and by the coupled eccentricity and ~~drag~~ attitude penalties that emerge during the evolution.

The second objective of this paper was methodological, and its outcome both reinforces the first ~~. An independent,~~ and qualifies it. The results of the original submission were produced by an implementation since found to be in error; every result has been regenerated with the open-source ~~re-implementation of the model reproduces~~ propagator validated in Section 9, which moved the regime from 700 to 800 km and reduced its magnitude without changing the mechanism. That propagator agrees with Orekit to 4×10^{-10} km in 14-day propagated mean semi-major axis for the J_2 and J_2-plus-drag dynamics; Orekit's N-plate model consumes the inertially fixed attitude directly and reproduces the full-model gain (6.13 km against 5.93 km here, the Sun ephemeris alone accounting for 0.5 km, or 9%), the sign of every common cell of the 800 km map, the ~~gain lobe, the eclipse-driven sign reversal of~~ 75-day transient and the ~~orbit-averaged SRP work, and the transient apogee-pumping-then-breakdown signature, demonstrating that the regime is a property of the dynamics and not of a single numerical implementation. Component-by-component cross-validation of that propagator against STK Astrogator shows excellent agreement for the conservative dynamics (~~eclipse-driven sign reversal. STK Astrogator reproduces the same behaviour once two configuration errors, found only by reading STK's effective inputs back, were corrected (a default 50 kg fuel load and a scalar-first quaternion): two-body and ~~J_2 agree to better than 0.05 km over 14 days)but a near order-of-magnitude divergence in the drag-induced decay, despite matching instantaneous drag force. This isolates atmospheric-drag modelling~~ J_2 agree to the metre, SRP alone to 0.1%, drag to 15% at 700 km and 5% at 800 km — ~~not the SRP mechanism~~ a residual accounted for by STK's co-rotating atmosphere and geodetic-altitude conventions — ~~as the dominant uncertainty in predicting where the passive-gain regime becomes net-positive,~~ and the eclipse-gated gain appears in both tools (+6.44 km in STK against +5.85 km here over 14 days). The near order-of-magnitude drag discrepancy of the original submission was therefore an artefact of the STK configuration, not a property of the tools. What remains uncertain is the thermosphere itself: replacing US-1976 by Harris–Priester or NRLMSISE-00 increases the 14-day drag decay at 800 km by factors of 3.1 and ~~it explains why the reproduced~~

~~regime is most clearly realised at higher altitude, where drag is weaker. A related practical finding is that industry N-plate SRP models are not readily configured for a passively, inertially fixed sail , which strengthens the case for purpose-built, inspectable, open-source tools when studying this class of problem~~1.6 (7.1 and 2.6 at 700 km), factors which straddle the factor-of-two density margin of the sensitivity analysis. Propagating the full model at the gain anchor with NRLMSISE-00 shows that the regime itself follows the solar cycle: at 800 km the 14-day gain is +13.2 km under low activity, +2.2 km under moderate activity and +0.5 km under the January 2026 conditions, and it vanishes under high activity, while at 700 km it exists only under low activity. The eclipse-gated force reduces the integrator to first-order convergence, but the 10 s production step lies within 0.2% of a DP853 reference, so the quantitative threshold of the regime is set by the thermospheric density model, not by the sail physics or by the integration.

Accordingly, the main contribution of this paper is twofold: scientifically, it establishes that a constrained transient gain regime exists at 800 km, that eclipse timing is its causal enabler, and that the same coupled dynamics that permit the early gain~~also~~ , together with the drift of a fixed sail normal away from the Sun, drive its later breakdown; and methodologically, it delivers and validates an open-source toolchain — released openly [48] and cross-checked against two independent propagators — for reproducibly studying passive-sail SRP–drag dynamics in LEO, which is increasingly the low-cost testbed of choice for sail technology. Because the quantitative gain threshold is ~~shown to be drag-model dependent~~set by the atmosphere model, future work should report ~~results~~ such thresholds together with the atmosphere model ~~used, tighten the analytical bounds for the gain window, reconcile the inter-tool drag discrepancy against in-orbit or higher-fidelity atmospheric data, and study whether limited attitude variation — informed by~~ and solar-activity level assumed; should examine limited attitude steering, re-pointing the spin axis every few weeks rather than controlling it continuously and informed by the optimal steering approaches ~~such as those~~ of Dachwald [34] and McInnes [14]~~— can delay the eccentricity- and drag-driven breakdown~~ , to counter the Sun–normal drift now identified as the dominant breakdown pathway without altering the fundamentally transient character of the effect; should tighten the analytical bounds on the gain window; and should ultimately be confronted with flight data from a passively spinning sail in a Sun-synchronous orbit.

CRediT authorship contribution statement

Ian Poon: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The open-source propagator, the Orekit cross-validation study, the STK Astrogator build and ~~sweep~~ probe scripts, the plate-definition files ~~, the cross-validation utilities and all reproduced figures~~ and the scripts that generate every figure and table are openly available in the project repository [48], as described in Section 9.8.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic) in order to edit and restructure the manuscript text, to review the simulation code, and to draft the response to the reviewers. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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